Population and Economic Growth When Malthus Meets Romer

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October 2026

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Abstract

Households disagree about whether a larger population raises living standards (Romerians) or lowers them (Malthusians). I embed this disagreement in a fertility model with dynastic altruism, cultural transmission, and semi-endogenous growth. Two forces sustain incorrect beliefs in the long run. First, differential fertility generates evolutionary selection, potentially favoring the wrong worldview. Second, aggregate fertility shapes the population path, which can render productivity observations uninformative. Together, these forces produce Malthusian cycles and Romerian information traps. In a calibrated illustration, inherited disagreement delays takeoff by centuries.

JEL Classification: D83, J13, O40, C72

Keywords: Fertility, heterogeneous beliefs, scale effects, learning, transition risk

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1 Introduction

A long-standing debate in economics concerns whether population growth improves living standards. Malthus (1798) emphasized the pressure that a growing population places on limited resources, with wages tending toward subsistence. Related concerns appear in Ehrlich (1968) and Meadows et al. (1972). Romer (1990) emphasized the contribution of a larger population to the production of ideas, whose benefits can extend throughout the economy. The importance of this channel, and the conditions under which it sustains growth, have been developed by Kremer (1993), Jones (1995, 2022), and Hopenhayn et al. (2022), among others.

The demographic changes of the past century give this debate particular relevance. World population rose from fewer than two billion to more than eight billion, while fertility subsequently fell below replacement across much of the developed world. The prospect of sustained population decline has brought renewed attention to the economic consequences of having fewer people (Jones, 2022; Doepke et al., 2023). A smaller population may ease pressure on resources and the environment, while also reducing the number of people producing new ideas. Households' views of these consequences matter for the demographic transition itself: parents deciding whether to have another child must assess the world that child is likely to inherit.

Rather than adjudicating between the two positions, this paper studies the implications of the disagreement itself for fertility and growth. The mechanism operates through parental altruism, cultural transmission, and learning. Parents who expect population growth to improve their descendants' prospects attach a different value to having children than parents who anticipate increasing population pressure. Differences in fertility change the representation of these worldviews in subsequent generations. Aggregate fertility also changes the population path and the productivity observations available for evaluating the competing views. Beliefs therefore influence both demographic adjustment and the evidence from which future generations learn. Figure 1 summarizes this feedback.

I develop an overlapping-generations model with dynastic preferences in the tradition of Barro and Becker (1989). Households share preferences and child-rearing costs but disagree about the sign of population's contribution to productivity growth. Romerians believe the contribution is positive; Malthusians believe it is negative. Productivity growth depends on the gap between population and a gradually adjusting reference population. This specification captures a semi-endogenous-growth property: a one-time population increase has a temporary growth effect, whereas sustained population growth can maintain one. Parents take aggregate transitions as given and choose fertility using their own forecasts of descendants' well-being.

I assume that these worldviews pass between generations through families and the wider social environment. Some children inherit their parents' beliefs. Others adopt a worldview according to a

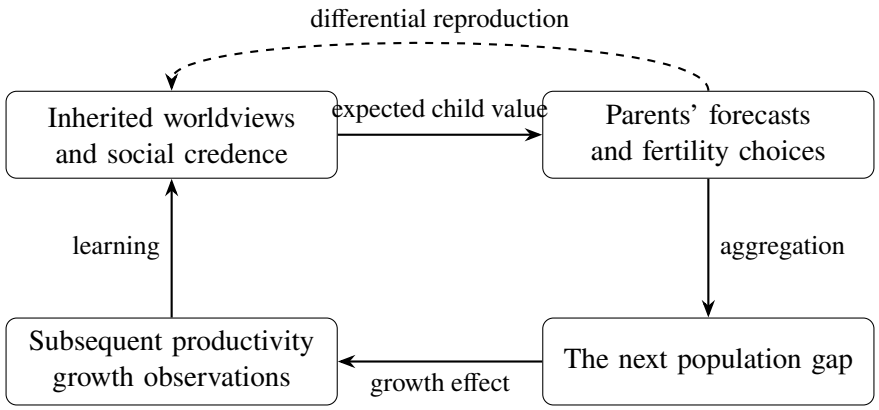


Figure 1. The Belief–Fertility–Growth Feedback

Notes: Fertility chosen today changes the next population gap, which affects a subsequent growth observation. The signal received during the current period depends on the predetermined current gap. The dashed arrow represents the effect of relative fertility on the composition of births.

shared assessment of aggregate growth evidence, which I call the social credence. This distinction between population composition and social credence allows reproductive selection and learning to proceed at different speeds. A worldview can become more common because its adherents have more children even as evidence reduces its social credence.

The household analysis establishes a state-dependent fertility ranking. Near the demographic reference, the fertility difference is proportional to a discounted expected path of the population gap. Romerians choose relatively more children when that discounted gap is positive, because they expect favorable population effects to improve their descendants' prospects. Malthusians choose relatively more children when it is negative, because they expect relief from population pressure. Along an exact reference path, both types forecast the same productivity distribution and choose the same fertility. The expected demographic trajectory thus determines which worldview gives parents the more favorable assessment of their children's future.

The aggregate analysis begins by separating demographic selection from learning. Selection favors whichever worldview is associated with higher fertility along the expected population path. Learning favors the model that better explains observed productivity growth. In a Malthusian world above the reference on balance, Romerians can gain representation through reproduction while growth observations support Malthusian beliefs. In a Romerian world below the reference on balance, Malthusians can enjoy the corresponding reproductive advantage while evidence favors Romerian beliefs. The conflict between selection and learning therefore depends on both the true technology and the expected demographic trajectory.

I next characterize the local fertility feedback and the information generated during demographic adjustment. Aggregate fertility can reinforce reference catch-up, slow it, or amplify an initial displacement. The same adjustment determines whether learning eventually resolves the

disagreement. At the reference, the rival models predict identical distributions of productivity growth. Nearby, the information in each observation is proportional to the squared population gap. The social credence identifies the truth when cumulative information diverges and converges to an interior limit when cumulative information is finite.

These results give rise to a dormant informational trap. When the fertility response returns the gap geometrically toward zero, the economy generates only finite information over the entire adjustment path. Fertility differences also disappear, leaving only finite cumulative demographic selection. Under a uniform local stability condition, a sufficiently small displacement returns toward the reference with high probability, and the social credence continues to assign positive probability to both models. With imperfect parental transmission, population composition gradually converges to that limiting credence. Households' fertility choices thus help restore an environment in which the disputed parameter has little influence on observable growth.

Beyond dormancy, the dynamics accommodate demographic reversals. A four-phase illustration in a Malthusian world traces expansion, correction, undershooting, and recovery, with each transition requiring the corresponding movements in fertility and composition. Appendix A.8 formalizes two additional properties of oscillatory adjustment: a local result establishes damped demographic cycles when the aggregate fertility response repeatedly takes the gap across the reference with diminishing amplitude, and such oscillations also generate only finite cumulative information. Under deterministic mean signals, the cumulative nature of learning rules out nonconstant periodic orbits with an interior social credence.

I then evaluate the economic significance of inherited disagreement through a correct-belief comparison. I consider two economies with the same true technology and the same initial demographic displacement. In one, households retain inherited disagreement; in the other, all households know the true scale effect. If inherited beliefs make a small positive displacement self-correcting while correct beliefs make it self-reinforcing, the informed economy reaches a growth-supporting region in finite time, while the economy with inherited disagreement can return to dormancy. This comparison identifies an entry delay attributable to beliefs. Sustained growth after entry depends on the subsequent equilibrium path.

The main application concerns delayed takeoff: beliefs inherited from a land-constrained past can weaken households' response to newly favorable scale effects, particularly near an uninformative demographic reference. A second application to modern fertility decline, developed in Section 6, shows how a larger Malthusian share lowers aggregate fertility while the discounted expected gap is positive and raises it once that gap becomes negative.

A calibrated illustration quantifies the delayed-takeoff mechanism. The calibration combines the paper's illustrative preferences with a strong Romerian scale effect in the population–technology feedback regime of Kremer (1993), a perceived Malthusian elasticity consistent with preindustrial

evidence (Lee, 1980; Clark, 2007), and vertical transmission in the range of cultural-transmission estimates (Bisin and Verdier, 2011). An economy in which only ten percent of households initially hold the correct Romerian worldview reaches the growth-supporting region 45 generations, roughly 1,125 years, later than an otherwise identical economy with correct beliefs. Below an initial Romerian share of about eight percent, takeoff fails to occur within two millennia of simulation: fertility choices return the population gap to its reference before evidence can accumulate, the quantitative counterpart of the dormant trap. Because takeoff requires the belief composition to cross a threshold, its timing is also highly dispersed across productivity histories, and a nontrivial share of simulated paths converges to dormancy despite a Romerian truth and an interior prior.

The final component of the paper assesses the empirical plausibility of the belief-to-fertility channel on which these results depend. Micro-level evidence from the General Social Survey supports this premise. Respondents who agree that the Earth cannot continue to support population growth report lower ideal and actual fertility. Conditional on age, education, sex, and survey year, the differences are approximately 0.27 and 0.29 children, respectively, or about 11 and 16 percent of the corresponding sample means. The association remains stable as marital status, race, religion, political orientation, and income are sequentially added, preserving roughly three-quarters of the baseline estimate; bias-adjusted bounds following Oster (2019) remain negative and economically meaningful. A decomposition into extensive and intensive margins attributes the actual-fertility relationship primarily to the decision to have any children. The association appears across education groups and survey waves, and remains present among respondents more likely to have completed childbearing. Because the survey waves were fielded against a salient and growing world population, a positive discounted expected gap is plausible in this context, and the observed gradient is consistent with the model's conditional prediction. Further variation across birth cohorts supports the state-dependent mechanism: the within-cohort Malthusian coefficient on ideal fertility is most negative for cohorts who reached prime childbearing age during the peak of world population growth, when the perceived gap was plausibly largest, and becomes less negative as growth subsequently declines.

The empirical differences are also economically meaningful. A back-of-the-envelope calculation holding the estimated fertility schedules fixed gives a difference of roughly 0.29 children between entirely non-Malthusian and entirely Malthusian populations, with a ten-percentage-point composition shift corresponding to about 0.03 children. These calculations illustrate the scale of the association rather than identify a causal aggregate effect. The survey does not observe respondents' demographic references or expected population paths, and its belief measure captures sustainability concerns rather than the scale-effect parameter alone. Complementary evidence from Robertson (2025) shows that officials of British India taught by Malthus at Haileybury subsequently provided less drought relief than officials taught by his successor and critic Richard Jones. That finding

supports the broader premise that economic ideas can durably affect consequential behavior, while the GSS speaks directly to the fertility margin.

Related Literature

The paper connects research on endogenous fertility and growth, cultural transmission, evolutionary selection, and learning. Its contribution is to make disagreement about population's economic consequences part of the process determining both demographic adjustment and the evidence available to resolve that disagreement.

The household problem builds on Becker (1960) and Barro and Becker (1989). Human-capital investment and differential fertility shape development in Galor and Weil (2000) and de la Croix and Doepke (2003). On population scale effects, Romer (1990), Kremer (1993), Jones (1995), and Hansen and Prescott (2002) distinguish channels through which population influences innovation and growth; Jones (2022) examines the consequences of population decline. The present paper takes the growth environment as given and studies how inherited beliefs affect households' demographic response to it. The correct-belief comparison isolates a transition delay attributable to beliefs rather than to the absence of favorable technological opportunities.

The transmission mechanism draws on Cavalli-Sforza and Feldman (1981) and Bisin and Verdier (2001, 2011). This literature distinguishes transmission within families from influence through the wider social environment and studies how cultural transmission interacts with reproductive selection. I take parental transmission as given rather than endogenizing socialization effort. The additional margin is recruitment through a social credence updated from aggregate growth observations. A worldview's prevalence and its evidentiary support can therefore move in different directions: higher fertility can increase its representation even while observations reduce its social credence. Fertility also changes the demographic path generating those observations.

The selection mechanism is closely related to evolutionary growth theory. Galor and Moav (2002) connect selection over preferences for child quality to the emergence of sustained growth. Galor and Michalopoulos (2012) show that the reproductive advantage of entrepreneurial traits can reverse across stages of development. State-dependent reproductive success is therefore not unique to the present framework. Here, selection operates on beliefs about an aggregate relationship rather than on preferences or productive traits. Its direction depends on the expected population path, and the reproductively successful belief need not be correct. Learning evaluates that belief against evidence generated by the resulting demographic adjustment.

The informational mechanism relates to self-confirming equilibrium (Fudenberg and Levine, 1993) and to equilibrium learning with potentially misspecified models (Esponda and Pouzo, 2016), where behavior determines which observations discipline beliefs. The present model retains standard Bayesian learning over a model class that includes the truth, but endogenous demographic

adjustment can leave only finite cumulative information about it. Moreover, atomistic parents do not choose fertility to experiment with the aggregate technology. Their privately optimal choices jointly determine whether subsequent generations encounter informative states. The contribution is to connect this information feedback to fertility and the intergenerational selection of worldviews.

More broadly, the paper relates to research on experience-based expectations (Malmendier and Nagel, 2011), information rigidity (Coibion and Gorodnichenko, 2015), and disagreement about long-run risks (Bakkensen et al., 2025). It places such disagreement on a demographic margin through which beliefs alter both future population composition and future evidence. The delayed-takeoff application also complements Mokyr (2002, 2016): inherited economic ideas can affect the response to growth opportunities before aggregate experience becomes sufficiently informative to settle the disagreement.

The remainder of the paper proceeds as follows. Section 2 presents the model and derives the fertility ranking. Section 3 analyzes selection, learning, local demographic feedback, dormancy, and demographic cycles. Section 4 develops the correct-belief comparison, the delayed-takeoff application, and a calibrated quantitative illustration. Section 5 provides the GSS evidence. Section 6 discusses an application to modern fertility decline, future research directions, and concludes. Proofs, additional results, and technical conditions appear in Appendix A; computational details and additional quantitative results appear in Appendix D.

2 Model

Time is discrete, $t = 0, 1, 2, \ldots$. Each generation of adults works, consumes, and chooses how many surviving children to raise. Those children become the next generation of adults and may inherit their parents' worldview. Households are atomistic and take aggregate transitions as given. Their only heterogeneity is belief about the economic effect of population; preferences and child-rearing costs are common.

2.1 Environment

Production, consumption, and the cost of children. A single final good is produced with effective labor, defined as adult time net of child-rearing, and the competitive wage equals productivity A_t. Raising children takes parental time: a parent who raises n children devotes a time share $\psi(n)$ to child-rearing and consumes $A_t(1 - \psi(n))$. The functional form of ψ, given in Appendix A.1, increases with n and absorbs the entire time endowment at a finite upper bound.

Fertility measures surviving adult descendants per current adult, with replacement normalized to one. The empirical outcomes measure children per survey respondent and retain their original units.

Population and its reference. Let ϕ_t denote the adult Romerian share. Aggregate fertility and the next adult population satisfy

$$\bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}, \qquad N_{t+1} = \bar{n}_t N_t. \tag{1}$$

Population affects growth through its distance from an adjusting reference. This population gap evolves according to

$$x_{t+1} = (1 - \mu) x_t + \ln \bar{n}_t, \qquad \mu \in (0, 1]. \tag{2}$$

Population growth widens the gap, while adjustment of the reference closes it. The parameter μ is the reference catch-up rate.

An accounting interpretation is $x_t = \ln N_t - \ln H_t$, where

$$\ln H_{t+1} - \ln H_t = \mu x_t.$$

The index H_t summarizes the scale against which population affects growth. Its interpretation allows for gradual adjustment of that scale without identifying it with a measured stock of ideas or a fixed carrying capacity.

The distinction between population and the population gap is important. A large population can have a small gap after the reference has caught up. The gap can also fall while population continues to grow if population growth is slower than reference adjustment. Convergence of the gap to zero can therefore occur at a population level different from its initial value.

With replacement fertility, the gap decays geometrically. With constant aggregate fertility, it converges to

$$x^{bgp} = \frac{\ln \bar{n}}{\mu}. \tag{3}$$

A one-time population increase consequently has a temporary gap effect, whereas sustained population growth can maintain a positive gap.

Productivity and disagreement. The baseline sets the common growth intercept to zero, $\bar{g} = 0$. Productivity follows

$$\ln A_{t+1} - \ln A_t = \theta^* x_t + \eta_{t+1}, \qquad \eta_{t+1} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \tag{4}$$

where

$$\theta^* \in \{\theta_R, \theta_M\}, \qquad \theta_R > 0 > \theta_M, \qquad \sigma > 0.$$

Romerians forecast using θ_R, and Malthusians forecast using θ_M. Actual growth is generated by the true parameter θ^*. Throughout, write

$$\Delta\theta = \theta_R - \theta_M > 0$$

for the belief gap, the magnitude of the disagreement.

In a Romerian world, population above the reference supports productivity growth. In a Malthusian world, the same gap depresses it. Below the reference, these growth effects reverse. Households agree on the observed gap and its adjustment law; they disagree about its implications for productivity.

This is a reduced-form semi-endogenous mechanism in the spirit of Jones (1995). Appendix B.1 discusses its relationship to conventional idea-production specifications. If aggregate fertility converges, conditional expected log-productivity growth converges to

$$\theta^* \frac{\ln \bar{n}_\infty}{\mu}. \tag{5}$$

At the reference, both models predict zero mean log-productivity growth and the same positive noise variance. A stationary reference path thus generates growth observations that are equally likely under the two worldviews. Productivity remains stochastic along this path.

Preferences. Parents value current consumption and their children's continuation utilities. Flow utility is CRRA, $u(c) = c^{1-\rho}/(1-\rho)$ with $\rho \in (0,1)$, and the weight placed on descendants is $\beta n^{1-\varepsilon}$, with $\beta \in (0,1)$ and $\varepsilon \in (0,1)$: the total weight on the next generation increases with the number of children at a diminishing rate.

The time-cost specification in Appendix A.1 is chosen so that flow utility net of the child cost is quadratic in n, with marginal-utility cost parameter $\gamma > 0$. The private choice problem is then strictly concave, and the fertility first-order condition takes the simple form (11) below. For illustration, a 15-percent time cost at replacement and $\rho = 0.5$ imply $\gamma \simeq 0.3122$. The analytical results retain the general preference parameters. Fertility lies in a compact interval $\mathcal{N}$ containing replacement, and discounted continuation values satisfy the finite-value restriction in Assumption 1.

CRRA utility and time costs make fertility invariant to the productivity level. A higher wage raises consumption opportunities and the earnings forgone during child-rearing in the same proportion. This structure isolates the effect of beliefs about descendants' future productivity from a direct current-income effect on fertility.

2.2 Beliefs, Learning, and Cultural Transmission

The adult Romerian share, ϕ_t, measures the prevalence of the worldview among current parents. The Romerian social credence, π_t, measures the weight assigned to it by aggregate evidence. Adults use their own worldview when choosing fertility and retain that type during the decision period. The social credence influences the worldview adopted by the next generation.

Learning from growth. The public signal is observed productivity growth,

$$s_t = \ln(A_{t+1}/A_t). \tag{6}$$

Under model j, the signal is Gaussian with mean $\theta_j x_t$ and variance σ^2. The social credence is updated by Bayes' rule given these two Gaussian likelihoods; the explicit formula is equation (A.18) in Appendix A.4. The initial credence is interior unless otherwise stated, allowing subsequent evidence to identify either model.

The KL divergence between the signal distributions is

$$\mathcal{I}_t = \frac{\Delta\theta^2 x_t^2}{2\sigma^2}. \tag{7}$$

This measures expected evidence in one observation. Information is zero at the reference and increases with the squared distance from it. A gap half as large supplies one quarter as much information per observation.

Transmission across generations. A child inherits the parent's worldview with probability $\lambda \in [0, 1]$. Otherwise, the child adopts the Romerian worldview with probability π_{t+1}, the updated social credence. Appendix equation (A.3) records the implied child-type transition probabilities.

Relative fertility determines the Romerian share among births. Parental transmission and social recruitment then determine the next adult share:

$$\phi_t^{birth} = \frac{\phi_t n_{R,t}}{\bar{n}_t}, \qquad \phi_{t+1} = \lambda\phi_t^{birth} + (1-\lambda)\pi_{t+1}. \tag{8}$$

With perfect parental transmission, differences in fertility govern the spread of worldviews, while social learning has no direct recruitment effect. With no parental transmission, the next generation's composition equals the updated social credence. Intermediate values allow families to preserve ideas while evidence gradually changes their prevalence.

2.3 Household Optimization and Equilibrium

Let $z = (x, \phi, \pi)$ denote the aggregate state relevant for normalized decisions. A type-j parent solves

$$V_j(A, z) = \max_{n \in \mathcal{N}} \Big\{ u\big(A(1 - \psi(n))\big) + \beta n^{1-\varepsilon}\, \mathbb{E}_j\big[V_{k'}(A', z')\big] \Big\}, \tag{9}$$

where the expectation is taken over the next growth observation under the parent's model j and over the worldview $k' \in \{R, M\}$ that the child adopts under the transmission rule above. A child who adopts a different worldview subsequently chooses and evaluates continuation utility according to that worldview.

Parents therefore account for the possibility that their descendants will hold different beliefs. The expected value of children reflects both the parent's forecast of the coming growth realization and the worldviews descendants may later adopt. Imperfect transmission gradually weakens the association between the initial parent's type and the outlook of distant descendants.

An individual household takes the aggregate transition as given. Its own fertility choice has no effect on the gap, social credence, or aggregate type shares. The aggregate feedback arises from the combined choices of all households.

Homogeneity gives

$$V_j(A, z) = A^{1-\rho} w_j(z). \tag{10}$$

Let $C_j(z)$ denote expected child continuation value normalized by current productivity; its full expression is in Appendix A.1. At an interior optimum,

$$\gamma n_j(z)^{1+\varepsilon} = \beta(1 - \varepsilon) C_j(z). \tag{11}$$

Higher expected child value raises fertility. Since preferences and costs are common, the fertility difference between worldviews comes entirely through their continuation values.

Definition 1 (Equilibrium)**.** A scale-invariant Markov equilibrium consists of normalized value functions w_R, w_M, fertility policies n_R, n_M, and aggregate transition rules such that:

(i) households solve (9), taking aggregate transitions as given;

(ii) individual policies aggregate according to (1);

(iii) population, the gap, social credence, and composition follow (1), (2), Bayes' rule (A.18), and (8);

(iv) each type forecasts future aggregate outcomes using these rules and its own productivity model.

Actual productivity growth is generated by θ^*.

The analysis follows a smooth equilibrium branch through the demographic reference. Assumption 1 ensures a well-defined household problem for a given aggregate rule. Assumption 2 supplies the differentiability and continuation conditions for the local equilibrium results. Global equilibrium existence and behavior outside the local region require additional analysis.

2.4 The Reference and the Fertility Ranking

Consider a population expected to remain exactly at its reference. Both worldviews then predict the same distribution of productivity at every future date. They assign the same value to children and choose the same fertility.

Lemma 1 (Common fertility at the reference)**.** *Under Assumption 1, a permanently zero-gap forecast gives both types the same unique optimal fertility n^0. A unique admissible value of β, given explicitly in equation (A.9) of Appendix A.2, yields $n^0 = 1$, provided this parameter choice satisfies Assumption 1.*

The baseline uses this replacement normalization: β is set so that both types choose replacement fertility at the reference. The normalization induces an effective reference discount factor $\delta \in (0, 1)$, which combines the altruism weight with the expected growth-adjusted utility scale at the reference; Appendix A.2 gives the formula.[1] At $\rho = \varepsilon = 0.5$, shock variance 0.04, and a 15-percent replacement time cost, $\delta \simeq 0.2530$. The common fertility choice is therefore consistent with a stationary reference path.

Nearby fertility choices depend on the future gaps parents expect. Let x^e_{t+k} denote a small common forecast perturbation around a *dormant path*—a path that remains at the reference with common composition and credence, $\phi = \pi = p$, analyzed in full in Section 3.4—and define

$$X^\lambda_t = \sum_{k=0}^{\infty} (\delta\lambda)^k x^e_{t+k}. \tag{12}$$

This is the discounted gap relevant for the fertility difference between parental types. Its weights reflect both discounting and the probability that descendants retain their initial parent's worldview. The corresponding sum discounted by δ alone, which governs the average fertility response, is introduced in Appendix A.3.

[1] Explicitly, δ equals the normalized altruism weight multiplied by $\exp\{\frac{1}{2}(1-\rho)^2\sigma^2\}$, the reference expectation of $(A'/A)^{1-\rho}$; see equation (A.7) in Appendix A.2.

Proposition 1 (State-dependent fertility ranking)**.** *Maintain Assumption 2. For a common forecast perturbation of size h around a dormant path with $\phi = \pi = p$,*

$$n_{R,t} - n_{M,t} = \frac{1-\rho}{1+\varepsilon}\,\Delta\theta\, X_t^{\lambda} + o(h). \tag{13}$$

Along perturbations with a nonzero first-order X_t^{λ}, Romerians have higher fertility when $X_t^{\lambda} > 0$, and Malthusians have higher fertility when $X_t^{\lambda} < 0$. On the exact zero-gap path, both choose fertility one.

For an above-reference expected path, Romerians anticipate favorable scale effects for their descendants, whereas Malthusians anticipate population pressure. These forecasts reverse for a below-reference path. The same parental concern for children's well-being therefore produces opposite fertility rankings in the two demographic regions.

Imperfect transmission reduces the influence of distant gaps on the fertility differential. A parent's worldview has less bearing on the continuation utility of descendants who are increasingly likely to adopt another outlook. Appendix A.3 gives the separate responses of both types, including their common response through average beliefs.

Current population conditions and expected future conditions can point in different directions. A declining population may remain above its reference for many generations; a growing population may remain below it. Equation (12) combines those expected gaps using the relevant intergenerational weights. If positive and negative gaps cancel, fertility differences vanish to first order. Exact equality follows when the entire expected gap path is zero.

3 Equilibrium Dynamics

Having established the household-level fertility ranking, this section traces where it leads in the aggregate. Relative fertility changes which families are represented in the next generation. Productivity observations change the social credence from which that generation may acquire its worldview. Their interaction determines whether population adjustment spreads a belief, corrects it, or leaves it unresolved.

3.1 Demographic Selection and Learning

Relative fertility makes the Romerian share among births differ from the adult share. From (8), this reproductive-selection margin equals $\phi_t(1-\phi_t)(n_{R,t} - n_{M,t})/\bar{n}_t$: at an interior share, it has the sign of the fertility differential.

Let $\mathbb{E}_t^{\theta^*}$ denote expectation under the true model conditional on public information before observing s_t. The current gap, fertility policies, and birth share are known at that time.

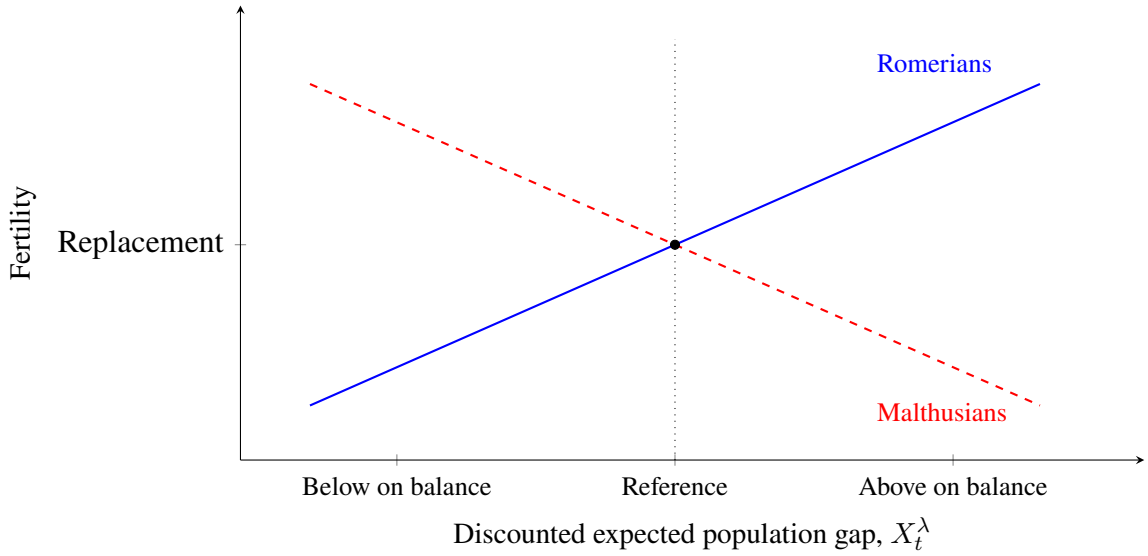


Figure 2. The State-Dependent Fertility Ranking

Notes: The figure illustrates Proposition 1, using symmetric scale-effect beliefs and perfect parental transmission for display. The slopes are illustrative. With imperfect transmission, the individual responses also contain a common component, as shown in Appendix A.3.

Proposition 2 (Selection and learning)**.** *The expected change in adult composition is*

$$\mathbb{E}_t^{\theta^*}[\phi_{t+1} - \phi_t] = \lambda \frac{\phi_t(1-\phi_t)(n_{R,t} - n_{M,t})}{\bar{n}_t} + (1-\lambda)(\pi_t - \phi_t) + (1-\lambda)\mathbb{E}_t^{\theta^*}[\pi_{t+1} - \pi_t]. \quad (14)$$

Under the local conditions of Proposition 1, the selection term has the first-order sign of X_t^λ. Expected posterior updating satisfies

$$\mathbb{E}_t^{\theta_R}[\pi_{t+1} - \pi_t] \geq 0, \qquad \mathbb{E}_t^{\theta_M}[\pi_{t+1} - \pi_t] \leq 0.$$

The three terms in (14) represent reproductive selection, recruitment toward the existing social credence, and recruitment induced by new evidence. The second term allows population composition to continue adjusting to information received in earlier periods.

In a Malthusian world with a positive discounted gap, Romerians have relatively more children while expected growth observations favor the Malthusian model. Their reproductive advantage can preserve Romerian representation even as its social credence falls. The speed of correction depends on both transmission and the gap between the existing population share and the credence.

Near a dormant state, a nonzero first-order fertility differential makes selection of order $|x_t|$. Expected posterior drift is of order x_t^2. The exact posterior expansions appear in Appendix A.4. This comparison concerns conditional means: the random component of posterior updating is also first-order in the gap, so a particular generation can experience a sizable belief revision. Selection

dominates learning near the reference, and learning dominates far from it. At the reference itself, both forces are exactly zero: disagreement is self-sustaining.

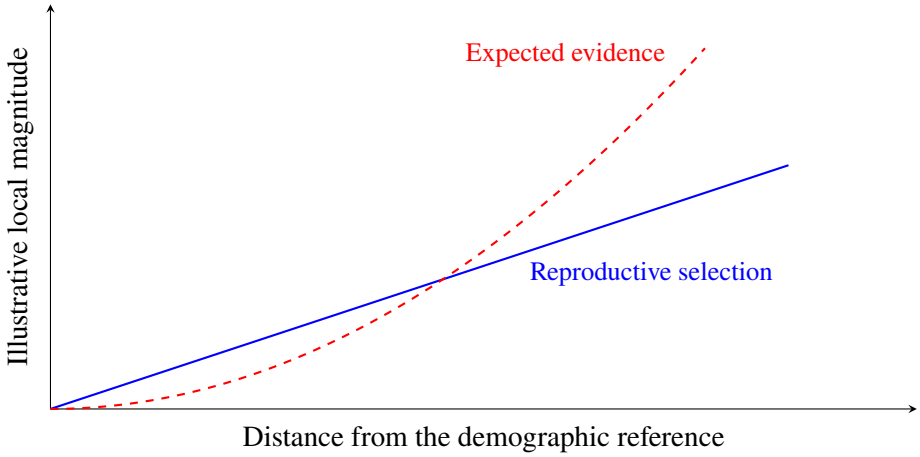


Figure 3. Local Orders of Selection and Information

Notes: Selection is generically first-order in the local gap, while expected information is second-order. Vertical scales are arbitrary. The figure describes local orders; a global crossover requires additional analysis. The random component of posterior updating is also first-order.

Table 1 summarizes the directions of selection and learning. Their conflict occurs above the reference on balance in a Malthusian world and below the reference on balance in a Romerian world.

Table 1. Selection and Learning Across Population Paths

Expected path	Malthusian world	Romerian world
Above reference	Selection: Romerians ×	Selection: Romerians ✓
	Learning: Malthusians ✓	Learning: Romerians ✓
Below reference	Selection: Malthusians ✓	Selection: Malthusians ×
	Learning: Malthusians ✓	Learning: Romerians ✓

3.2 Local Demographic Feedback

A population displacement changes fertility through parents' forecasts of the future gap. Those fertility changes, in turn, affect the persistence of the displacement. The local equilibrium response accounts for this entire continuation.

At a dormant state $(x, \phi, \pi) = (0, p, p)$, define the local gap coefficient

$$a(p) = 1 - \mu + \left. \frac{\partial \ln \bar{n}(x, \phi, \pi)}{\partial x} \right|_{(0,p,p)}, \tag{15}$$

so that

$$x_{t+1} = a(p)x_t + o(|x_t|). \tag{16}$$

The derivative term is the aggregate log-fertility response to the gap; $a(p)$ combines that response with reference catch-up. For compactness, let

$$K = \frac{1-\rho}{1+\varepsilon}, \qquad \bar{\theta}_p = p\theta_R + (1-p)\theta_M. \tag{17}$$

Proposition 3 (Local equilibrium response)**.** *Under Assumption 2, suppose* $\delta|a(p)| < 1$*. Then*

$$\big(a(p) - (1-\mu)\big)\big(1 - \delta a(p)\big) = K\bar{\theta}_p, \tag{18}$$

and

$$n_R - n_M = \frac{K\Delta\theta}{1 - \delta\lambda a(p)}x + o(|x|). \tag{19}$$

On the real, nondegenerate root connected to $a = 1-\mu$ *at* $\bar{\theta}_p = 0$*, the coefficient* $a(p)$ *increases with* $\bar{\theta}_p$*. The root and individual policy slopes are given in Appendix A.5.*

A negative aggregate fertility response reinforces reference catch-up. Above the reference, a sufficiently Malthusian average outlook reduces fertility because parents anticipate population pressure. Below the reference, it raises fertility because parents anticipate relief. This response supplies a restoring force.

A more Romerian outlook raises the local gap coefficient. From a positive coefficient below one, it slows convergence and can eventually produce amplification. From a negative coefficient, it reduces the extent of overshooting. The direction of the change in $a(p)$ is unambiguous, while its effect on convergence speed depends on the starting coefficient.

Strict local contraction requires

$$-1 < a(p) < 1; \tag{20}$$

equivalently, the aggregate fertility response $a(p) - (1-\mu)$ lies between $-(2-\mu)$ and μ. A coefficient between zero and one produces a same-side return. A coefficient between minus one and zero produces alternating, diminishing displacements. A coefficient above one amplifies a positive displacement. The lower stability bound restricts the strength of corrective fertility responses.

At the reference, both types choose fertility one. A small change in their population weights therefore has no direct first-order effect on aggregate fertility. Its fertility effect enters through the interaction with a nonzero gap. This property permits the local gap coefficient to be evaluated at the initial belief distribution even though belief innovations themselves can be first-order.

3.3 Information Accumulation

The informativeness of the realized population path determines whether learning eventually resolves the disagreement. Define cumulative signal information by

$$\mathcal{C}_T = \sum_{t=0}^{T-1} \mathcal{I}_t. \tag{21}$$

Since each $\mathcal{I}_t$ is proportional to x_t^2, learning depends on the cumulative squared gap.

Theorem 1 (Information and posterior convergence)**.** *Suppose* $\pi_0 \in (0,1)$*. Under either true model:*

(i) On $\{\sum_{t=0}^{\infty} \mathcal{I}_t = \infty\}$*, the posterior identifies the truth almost surely:*

$$\pi_t \to 1 \quad \text{if } \theta^* = \theta_R, \qquad \pi_t \to 0 \quad \text{if } \theta^* = \theta_M.$$

(ii) On $\{\sum_{t=0}^{\infty} \mathcal{I}_t < \infty\}$*, the posterior converges almost surely to an interior limit* $\pi_\infty \in (0,1)$*.*

Persistent fertility above or below replacement keeps the limiting gap away from zero and supplies infinite information. A geometric return to the reference supplies only finite information. Slower convergence can still yield identification if the squared gaps have a divergent sum.

Gap reversals can continue to support the same true model. In a Malthusian world, weak growth above the reference and stronger growth below it both favor the Malthusian explanation. The sign of the gap changes the model's growth prediction, while the evidence accumulates in support of the same parameter.

Identification by the social credence can precede the spread of correct beliefs throughout the population. Parental transmission can sustain an incorrect type, particularly when that type has a fertility advantage. Appendix Corollary 1 shows that the correct population share converges to one if transmission is imperfect and correct believers eventually have fertility at least as high as incorrect believers. With perfect transmission, an eventually uniform fertility advantage and a positive initial correct share are sufficient.

3.4 The Dormant Informational Trap

A stable fertility response can bring the economy toward a state where the rival growth models are observationally identical. Exactly at the reference, equal fertility preserves the gap and leaves the social credence unchanged. Nearby, geometric convergence makes the remaining sequence of observations only finitely informative.

Write the equilibrium gap map as

$$F(x, \phi, \pi) = (1 - \mu)x + \ln \bar{n}(x, \phi, \pi).$$

Assumption 3 formalizes uniform local stability. It requires $|F_x(0, \phi, \pi)| \leq q_0 < 1$ throughout a belief neighborhood $[p - d, p + d]^2 \subset (0, 1)^2$. This permits beliefs to fluctuate while preserving a contracting gap response.

Theorem 2 (Dormancy and persistent disagreement)**.** *Maintain the replacement normalization of Lemma 1 and the smooth dormant equilibrium branch.*

(i) If $x_0 = 0$, then, along the maintained dormant branch,

$$x_t = 0, \qquad n_{R,t} = n_{M,t} = 1, \qquad \pi_t = \pi_0, \qquad \phi_t = \pi_0 + \lambda^t(\phi_0 - \pi_0).$$

(ii) Suppose Assumption 3 holds and $\phi_0 = \pi_0 = p$. For any $q \in (q_0, 1)$ and sufficiently small $|x_0|$, there are a constant $C < \infty$ and an event E such that

$$\mathbb{P}^{\theta^*}(E) \geq 1 - \frac{C|x_0|}{d}, \qquad |x_t| \leq q^t|x_0| \quad \text{on } E.$$

On E, cumulative information and absolute reproductive selection are finite, and π_t converges to an interior limit. Population converges to a positive finite level, and long-run average log-productivity growth is zero.

The nearby result allows unusual growth observations to shift beliefs outside the stable neighborhood. Its probability bound shows that such exit becomes unlikely as the initial gap approaches zero. Appendix A.7 also bounds cumulative information and the movement of the limiting credence. On the no-exit event, adult composition converges to the limiting credence when transmission is imperfect; with perfect transmission, composition converges to a possibly different limit.

The baseline's shock structure is important. Productivity shocks change the productivity level without directly displacing the gap. At the reference, they have the same likelihood under both models and therefore leave the social credence unchanged. Homogeneity also removes their direct level effect on fertility. Together, these properties allow the exact reference path to persist.

Productivity continues to fluctuate on dormant paths. The zero-intercept baseline implies a zero long-run average log-growth rate after the population contribution has disappeared. Arithmetic expected growth can remain positive because of multiplicative noise.

The theorem describes a joint demographic and informational outcome. Household behavior restores population toward its reference while leaving the scale-effect disagreement unresolved.

Whether that persistence delays a growth opportunity depends on the correct-belief comparison in Section 4.

3.5 Adjustment in Malthusian and Romerian Worlds

In a Malthusian world, the fertility ranking and the direction of learning create a natural expansion–correction sequence. Above the reference on balance, Romerians have the reproductive advantage while the data favor Malthusians. Below the reference on balance, Malthusians have the reproductive advantage and the data continue to favor their model.

The following four-phase illustration traces how these forces can interact. Each transition requires the corresponding aggregate fertility and composition movements. A formal local result then establishes diminishing overshoots under a negative stable gap response.

A Four-Phase Malthusian Cycle

Consider a Malthusian economy with a small positive gap and a positive discounted expected gap. The illustration assumes $0 < \lambda < 1$, so both reproductive transmission and recruitment through social evidence influence population composition. An initial displacement places the economy away from the exact dormant path.

Phase I: Expansion and reproductive selection. Romerians anticipate favorable prospects along the above-reference path and choose relatively more children. Their birth share consequently exceeds their adult share. If this reproductive effect is large enough relative to recruitment toward the social credence, the adult Romerian share rises. If aggregate fertility also exceeds the rate needed to offset reference catch-up, the population gap widens.

This phase therefore requires two net movements: increasing Romerian representation and an increasing gap. Relative Romerian fertility supplies the selection component, while aggregate fertility determines the demographic movement. In this region, reproduction can spread the incorrect worldview even as expected growth observations favor the true Malthusian model.

Phase II: Learning and demographic correction. A positive gap depresses expected growth in the Malthusian environment. Accumulating observations tend to lower the Romerian social credence. Wider gaps make the rival predictions easier to distinguish, increasing the information supplied by each observation.

Demographic correction begins when the combined effects of learning, recruitment, and fertility reduce the gap. Parental transmission can delay this response because the adult population continues to contain families whose worldview has lost evidentiary support. The timing depends on the full

composition law and the fertility policies; the local order comparison alone leaves the turning point undetermined.

During the initial correction, population can continue to grow while the gap narrows. This occurs when fertility remains above replacement but falls below the rate required to keep population ahead of its reference. Population itself begins to decline once fertility falls below replacement.

Phase III: Undershooting and recovery. A sufficiently strong correction carries the gap below zero. Once the discounted expected path is also negative, Malthusians choose relatively more children. They anticipate that reduced population pressure will improve their descendants' economic prospects, while Romerians anticipate weaker idea production.

Malthusian beliefs thus support fertility restraint above the reference and relatively higher fertility below it. In the below-reference phase, selection and expected learning both favor Malthusians. Their fertility response can support recovery if aggregate fertility rises enough to close the negative gap.

Entry into this phase requires an actual crossing of the reference and a negative discounted expected gap. Paths that correct more gradually can approach the reference from above.

Phase IV: Damping and possible dormancy. As the gap narrows, fertility differences decline and new observations become less informative. A stable aggregate response brings the economy toward the reference. A stable negative response produces further crossings with diminishing amplitude.

The social credence retains the evidence accumulated during earlier phases, so later demographic waves occur with a different belief history. The economy can eventually approach dormancy with an interior limiting credence if the remaining gaps generate only finite information.

The changing fertility ranking gives Malthusian beliefs a role in both correction and recovery. Expected population pressure lowers the value of children along the above-reference path; expected relief raises it relative to Romerian beliefs along the below-reference path. The strength and persistence of this response determine whether adjustment remains on one side of the reference or proceeds through overshooting.

Appendix A.8 formalizes two additional properties of oscillatory adjustment. Corollary 2 shows that a uniformly negative local gap response produces alternating, geometrically diminishing displacements that generate only finite cumulative information, extending the dormancy conclusion of Theorem 2 to damped demographic cycles. Proposition 5 shows that, under deterministic mean signals, the system admits no nonconstant periodic orbit with an interior social credence, ruling out permanent belief-driven oscillations. Recurrent fixed-amplitude disagreement cycles would require additional forces that renew uncertainty or reduce the weight on past evidence.

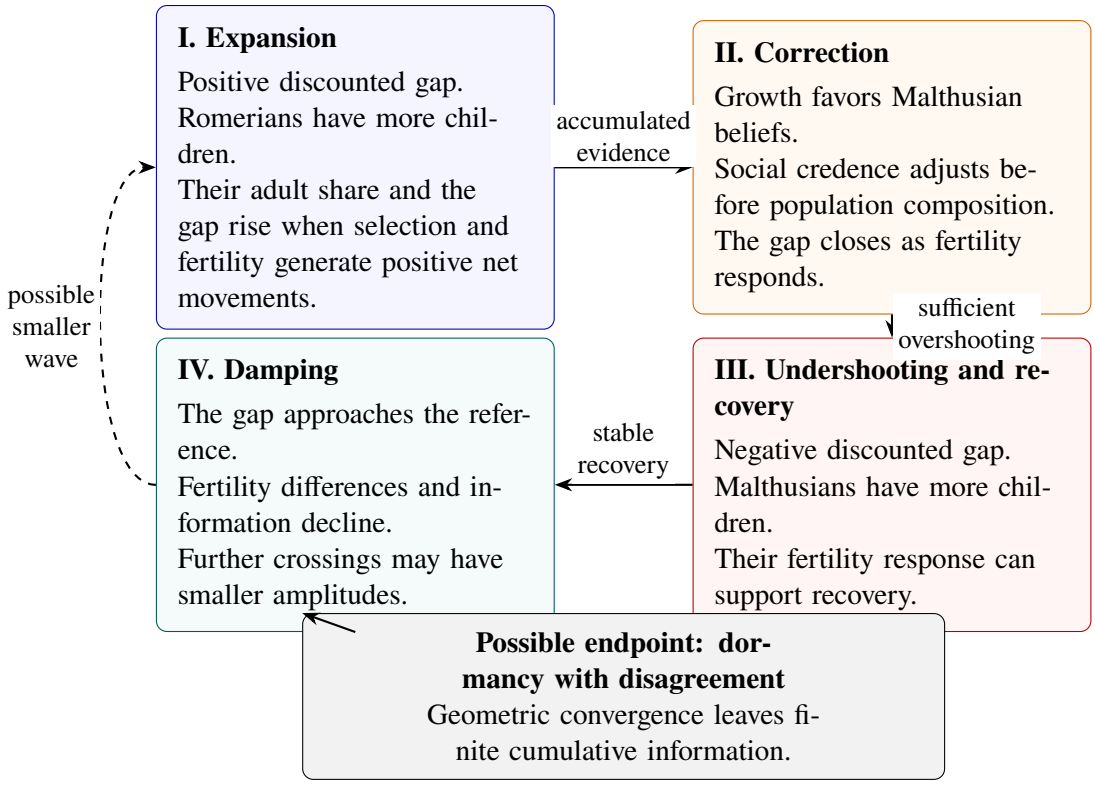


Figure 4. A Four-Phase Illustration of Malthusian Demographic Adjustment

Notes: The diagram organizes a conditional expansion–correction sequence. Each arrow requires the associated movements in fertility, composition, and the discounted expected gap. Appendix A.8 gives exact phase diagnostics. The dashed arrow represents further demographic overshooting with accumulated belief history. Appendix Corollary 2 establishes local diminishing sign reversals under a uniformly negative stable gap response.

Adjustment in a Romerian World

Above the reference on balance, a Romerian world aligns selection and learning. Romerians have relatively more children, and expected growth observations support their model. The aggregate trajectory still depends on the equilibrium fertility response and its behavior beyond the local region.

Below the reference on balance, Malthusians can gain representation through reproduction while observations favor Romerian beliefs. Their expectation of relief from population pressure supports relatively higher fertility, even though the true growth effect of the negative gap is unfavorable. This configuration is relevant to demographic adjustment during fertility decline.

3.6 Scope of the Mechanism

The informational trap is most relevant when the economy approaches a state where growth predictions and fertility choices become similar. Its persistence also depends on the sources of information and demographic variation available outside the baseline feedback.

Regional comparisons or outside research can identify the scale effect while the domestic

gap remains small. Persistent demographic disturbances can repeatedly move the economy into informative states. Type-specific preferences for children can preserve reproductive selection even when growth forecasts coincide.

These extensions affect the distinction between exact dormancy and a long adjustment delay. The baseline isolates information generated by the economy's own productivity observations along its endogenous demographic path. Appendix B.5 discusses the corresponding extensions.

4 Quantitative Application: Delayed Takeoff

The economic consequences of inherited disagreement depend on the response that correct beliefs would produce under the same opportunities. This section constructs that comparison, applies it to historical takeoff, and quantifies its implications. The central question is whether the initial share of Romerians in a population can determine the difference between a stagnant and a growing world.

4.1 A Correct-Belief Benchmark

Consider two economies with the same true technology, preferences, population-adjustment process, initial productivity and population, and exogenous productivity innovations. Economy H begins with inherited disagreement. Economy C begins with every household knowing the true scale effect. Both receive the same initial population displacement.

Suppose the truth is Romerian. In economy C, $\phi = \pi = 1$ at every date. For a positive threshold h, define the first entry time

$$T_h^k = \inf\{t \geq 0 : x_t^k \geq h\}, \qquad k \in \{H, C\}, \tag{22}$$

with $T_h^k = \infty$ if entry never occurs. Reaching h gives conditional expected log-productivity growth of at least $\theta_R h > 0$.

This entry criterion measures the response to an initial growth opportunity. Sustained growth after entry depends on the subsequent equilibrium path. Appendix A.9 also defines the excess probability of delayed entry and the difference in truncated expected entry times.

Proposition 4 (Belief-mediated entry delay)**.** *Suppose $\theta^* = \theta_R$. Economy H satisfies Assumption 3 and starts from $\phi_0 = \pi_0 = p$. Economy C has a differentiable dormant gap map F_C with*

$$F_C(0) = 0, \qquad F_C'(0) = a_C > 1.$$

For a sufficiently small threshold $h > 0$ and common initial gap $x_0 = z \in (0, h)$,

$$T_h^C < \infty, \qquad \mathbb{P}^{\theta_R}(T_h^H = \infty) \geq 1 - \frac{Cz}{d}.$$

For any $a_- \in (1, a_C)$, the threshold can be chosen so that

$$T_h^C \leq \left\lceil \frac{\ln(h/z)}{\ln a_-} \right\rceil.$$

The two economies respond differently to the same displacement. Correct beliefs make the positive gap sufficiently favorable to encourage fertility that enlarges it. Inherited beliefs instead produce a return toward the reference. That return also reduces the information available for revising those beliefs.

Both economies can remain at the exact reference when they start there. The proposition holds the initiating displacement fixed and compares their responses to it. As the displacement becomes smaller, the inherited-belief economy is increasingly likely to remain in its stable neighborhood, while the informed economy still reaches the fixed local threshold in finite time.

The proposition measures delayed entry into a growth-supporting region. A welfare assessment would additionally require accounting for child-rearing costs, population, and dynastic utility. The comparison attributes the entry difference to inherited beliefs by holding technology and the initiating disturbance constant.

4.2 The Delayed-Takeoff Application

The delayed-takeoff application considers an economy whose scale effect has become favorable while its households retain beliefs formed in an earlier environment. In terms of the model, the question has three dimensions: How long could initial pessimism delay growth takeoff? How large are the welfare consequences of imperfect information? How large is the role of luck or cultural entrepreneurship (Mokyr, 2016)?

Unified growth accounts explain changes in the opportunities facing households and firms through technological transformation and human-capital demand (Hansen and Prescott, 2002; Galor and Weil, 2000; Galor, 2011). The present mechanism concerns households' demographic response once favorable scale effects are available. It takes the post-transition technology as given and represents the inherited past through the initial distribution of worldviews.

A land-constrained history can support Malthusian beliefs. Historical episodes linking lower population to improved living standards provide motivation for such an interpretation (Clark, 2007; Pamuk, 2007). Within the model's Malthusian environment, a negative population gap raises expected log-productivity growth, and the resulting evidence favors the Malthusian model.

After the scale effect changes, households learn about the fixed post-transition parameter. Near the demographic reference, the rival post-transition models make similar predictions. Inherited beliefs can consequently remain consistent with the observed growth experience for a long time.

If those beliefs also stabilize the reference, their persistence affects the response to demographic opportunities. A small positive displacement produces a limited fertility response and a return toward the reference. The economy then supplies little additional evidence about the newly favorable scale effect. The availability of the technology and the demographic response to it can therefore be separated in time.

Proposition 4 identifies the condition under which this separation is attributable to beliefs: correct beliefs would amplify the common positive displacement, while inherited beliefs contract it. The comparison links disagreement to an entry delay through both fertility and endogenous information.

A change in cultural outlook can alter this response before aggregate growth experience becomes decisive. A more favorable interpretation of population changes the fertility response to future demographic displacements, potentially bringing the economy into states where competing models are easier to distinguish. This channel is related to the role of ideas and attitudes toward improvement emphasized by Mokyr (2002, 2016).

Other departures are also possible. A sufficiently large demographic displacement may carry the economy beyond the local return region, and outside evidence may identify the new scale effect directly. The model isolates the timing margin associated with inherited population beliefs while leaving the historical sources of technological change, education, institutions, and cultural transformation to the broader accounts.

4.3 A Quantitative Illustration

The results so far establish signs and local orders. This subsection asks whether the mechanism can matter at historically relevant magnitudes. I simulate the equilibrium under a calibration in which the truth is Romerian and the scale effect is strong enough that correct beliefs would amplify a demographic displacement, so the premise of Proposition 4 holds. A model period is a generation of 25 years. Fertility is reported in total-fertility-rate units, with replacement fertility $n = 1$ corresponding to a TFR of two and the upper bound of $\mathcal{N}$ to a TFR of six. Appendix D contains computational details, the welfare construction, and additional robustness results.

Calibration. Table 2 lists the parameters. The preference block carries over the illustrative values of Section 2: $\rho = \varepsilon = 0.5$, a 15 percent child-rearing time cost at replacement pinning γ, and the replacement normalization $\beta = \beta^{\dagger}$ of Lemma 1, which together imply $\delta \simeq 0.253$. The dynastic curvature $\varepsilon = 0.5$ is consistent with the restrictions emphasized for Barro–Becker models by Jones

and Schoonbroodt (2010). The shock volatility $\sigma = 0.2$ per generation corresponds to an annualized standard deviation of about four percent for the random-walk component of productivity.

Three parameters govern technology and transmission. The reference catch-up rate $\mu = 0.10$ implies that, under replacement fertility, a displacement halves roughly every 165 years, in line with the very slow transition dynamics of semi-endogenous growth frameworks (Jones, 1995, 2022). The Romerian scale effect $\theta_R = 0.5$ delivers a long-run level multiplier $\theta_R/\mu = 5$, placing the economy in the strong population–technology feedback regime that motivates the very-long-run evidence in Kremer (1993); it also implies expected growth of about two percent per year once the gap reaches the entry threshold $h = 1$. The Malthusian parameter $\theta_M = -0.15$ corresponds to a perceived long-run elasticity of living standards with respect to population of $-\theta_M/\mu = 1.5$, consistent with the preindustrial English estimates of roughly -1.5 in Lee (1980) and within the 0.55–1.6 range reported by Clark (2007). The vertical transmission probability $\lambda = 0.6$ lies in the middle of the parent–child attitude correlations of roughly 0.4 to 0.7 emphasized in the cultural-transmission literature (Bisin and Verdier, 2001, 2011; Fernández et al., 2004); values of 0.4 and 0.8 are examined in the appendix. The initial displacement is $x_0 = 0.30$.

These values imply $\bar{\theta}^{dest} \simeq 0.224$ from (A.29). Because θ_R exceeds this threshold, the correct-belief economy amplifies a positive displacement ($a_C \simeq 1.13$), while amplification in the inherited-disagreement economy requires the average belief to cross $\bar{\theta}^{dest}$, corresponding to a Romerian share of $p^* \simeq 0.58$. The quantitative question is whether, and how quickly, selection and learning carry the composition across that threshold.

Table 2. Baseline Calibration

Parameter	Value	Target or source
ρ	0.50	Illustrative curvature, Section 2
ε	0.50	Dynastic curvature (Barro and Becker, 1989; Jones and Schoonbroodt, 2010)
γ	0.312	15% child-rearing time at replacement (Appendix A.1)
β	0.252	Replacement normalization $\beta^\dagger$ (Lemma 1)
σ	0.20	$\approx 4\%$ annual s.d. of productivity innovations
μ	0.10	Gap half-life ≈ 165 years (Jones, 1995, 2022)
θ_R	0.50	Long-run multiplier $\theta_R/\mu = 5$ (Kremer, 1993); 2%/yr growth at entry
θ_M	-0.15	Perceived level elasticity $-\theta_M/\mu = 1.5$ (Lee, 1980; Clark, 2007)
λ	0.60	Vertical transmission (Bisin and Verdier, 2001, 2011)
x_0	0.30	Initial displacement
h	1.00	Entry threshold in (22)

Notes: A period is 25 years and the truth is Romerian, $\theta^* = \theta_R$. Implied constants: $\delta = 0.253$, $K = 1/3$, $\bar{\theta}^{dest} = 0.224$, $p^* = 0.576$, and $a_C = 1.134$. Fertility is restricted to a TFR range of 0.6 to 6 (Appendix D.1). Appendix D discusses provenance in detail, including the remark that modern semi-endogenous estimates imply a scale effect below $\bar{\theta}^{dest}$.

How long could initial pessimism delay growth takeoff? Figure 5 traces three economies that differ only in the initial Romerian share, $\phi_0 = \pi_0 \in \{0.02, 0.10, 0.30\}$, each starting from the same displacement; paths are computed under the mean signal sequence, with sampling uncertainty deferred to Figure 7. The three initial compositions produce three qualitatively different histories. With $\phi_0 = 0.30$, evidence identifies the Romerian model within about eight generations, the adult share crosses p^* almost immediately, and the gap reaches the entry threshold at generation 20, after 500 years. With the benchmark $\phi_0 = 0.10$, the mostly Malthusian fertility response first contracts the gap from 0.30 to a minimum of 0.09 around generation 20, almost exactly when selection and learning carry the composition across p^*; only then does the gap begin to compound, with takeoff at generation 55, after roughly 1,375 years. During the first five centuries the benchmark economy's TFR stays between 2.01 and 2.02, a near-stagnant path that would be hard to distinguish from dormancy in the data it generates. With $\phi_0 = 0.02$, the credence converges to an interior limit near 0.10, the composition converges to that credence, and the gap and fertility return to the reference and replacement: the dormant trap of Theorem 2, realized despite a Romerian truth and an interior prior.

The initial Romerian share thus separates a stagnant world from a growing one. Under the same true technology and the same initial displacement, one economy takes off within 500 years, another takes over a millennium, and a third never escapes. The only difference is the inherited belief composition.

Figure 6 reports the takeoff delay as a function of initial beliefs, measured relative to the correct-belief economy, which reaches the threshold in 10 generations, or 250 years. For $\phi_0 \leq 0.07$ the economy fails to take off within the 2,000-year simulation horizon; this censored region is the numerical counterpart of the trap. Just outside it the delay is extreme: 1,675 years at $\phi_0 = 0.09$ and 1,125 years at the benchmark $\phi_0 = 0.10$. The delay then falls quickly, below 300 years for ϕ_0 above one quarter and below 100 years for ϕ_0 above 0.55. The curve's convexity near the trap boundary reflects the model's central feedback: where beliefs are most pessimistic, the fertility response keeps the gap smallest, so the observations that could correct those beliefs are least informative.

How large are the welfare consequences? To attach a welfare number, I compute the uniform proportional consumption supplement that makes the initial dynasty in the disagreement economy indifferent to the correct-belief path over the first 20 generations, before the fertility cap binds in either economy; Appendix D.2 gives the construction. The supplement is $\omega \simeq 0.74$ percent of consumption per generation, and it is stable, between 0.71 and 0.75 percent, across the $\lambda \times \sigma$ grid in Appendix Table A.11. Two forces keep this number modest despite the millennium of delay. Fewer children in the disagreement economy mean slightly higher consumption in the first generations, and the effective dynastic discount factor of $\delta \simeq 0.25$ per generation places little weight on the

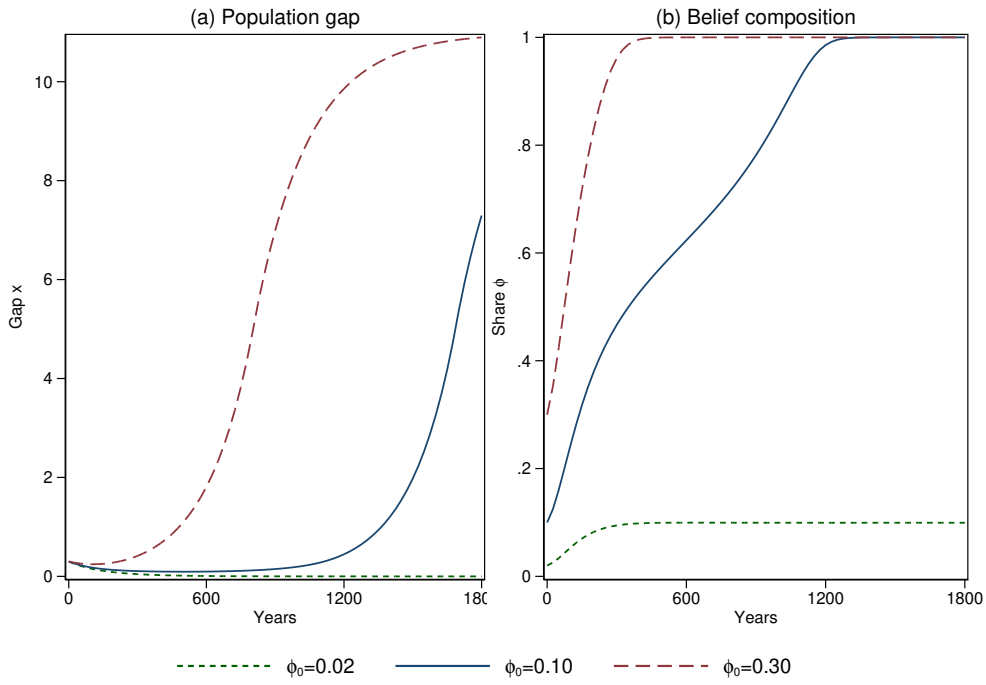


Figure 5. Threshold Dynamics Across Initial Belief Compositions

Notes: Mean-signal equilibrium paths under the Romerian truth for three initial Romerian shares, with $x_0 = 0.30$ and $\phi_0 = \pi_0$ in each case. A generation is 25 years. In panel (c), replacement corresponds to a TFR of two and the upper bound of $\mathcal{N}$ to a TFR of six; paths at the bound reflect the cap. The amplification threshold is $p^* \simeq 0.58$.

distant future where the consumption differences become enormous (Appendix Figure A.8). The measure is therefore a conservative dynastic valuation of the delay.

How large is the role of luck? The deterministic paths understate the role of luck. Figure 7 reports pointwise 5th, 50th, and 95th percentiles across 500 Monte Carlo replications of the productivity-shock sequence for the benchmark economy with $\phi_0 = \pi_0 = 0.10$. The dispersion in takeoff timing is large and asymmetric. The fastest five percent of paths reach the threshold within about 19 generations, under 500 years, because favorable early growth realizations raise the credence while the initial gap is still large and informative. The median path crosses only around generation 68, later than the mean-signal path, reflecting the nonlinearity of the threshold dynamics: adverse early realizations lower the credence precisely when the contracting gap is about to shut off the flow of information. The bottom five percent of paths never escape: their gap converges to zero and their Romerian share settles near 0.03, an interior limit of the credence in the sense of Theorem 1(ii). The trap therefore arises with positive probability under the Romerian truth, beyond the limiting case of small displacements.

This wide dispersion suggests a substantial role for what Mokyr (2016) calls cultural en-

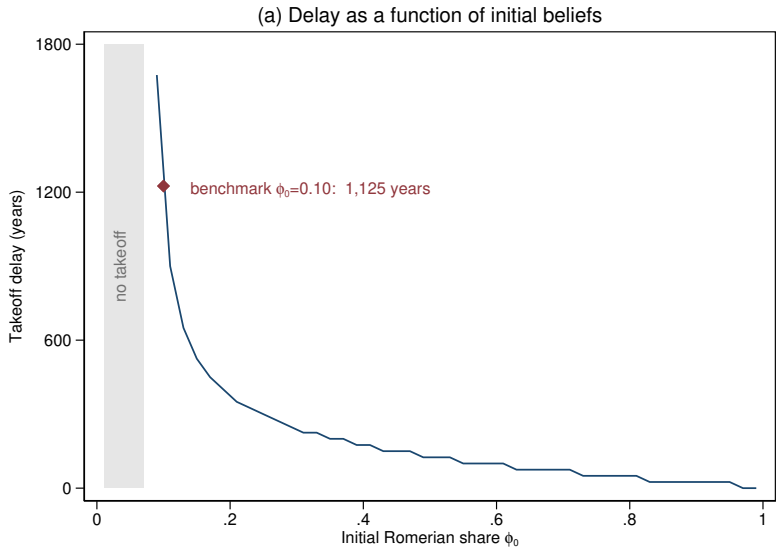


Figure 6. Takeoff Delay Relative to the Correct-Belief Economy

Notes: Delay in reaching the entry threshold $h = 1$ relative to the correct-belief economy (250 years), as a function of the initial Romerian share. The shaded region marks no takeoff within the 2,000-year simulation horizon. The diamond marks the benchmark $\phi_0 = 0.10$. Appendix Figure A.6 displays the population-gap paths for the inherited-disagreement and correct-belief economies of Proposition 4.

trepreneurship. A society that happens to encounter favorable early growth shocks—or one in which intellectual leaders successfully promote a more Romerian outlook—can cross the amplification threshold centuries earlier. The mechanism does not require that cultural entrepreneurs know the true model. It only requires that they shift the population's outlook far enough for the resulting fertility response to bring the economy into states where the data can do the rest.

Sensitivity. Parameter sensitivity (Appendix Table A.10) follows the mechanism's logic. The delay rises with parental transmission, from 950 years at $\lambda = 0.4$ to beyond the horizon at $\lambda = 0.8$, because stronger family transmission slows recruitment toward the improving credence. It falls sharply with less noise, to 175 years at $\sigma = 0.1$, and becomes censored at $\sigma = 0.3$, since information per observation scales with $1/\sigma^2$. Halving or doubling the Malthusian pessimism moves the delay only modestly, between 975 and 1,225 years, because a larger $\Delta\theta$ strengthens the corrective fertility response and speeds learning in offsetting directions. The strong scale effect is a deliberate choice for the historical takeoff question: as discussed in Appendix D, mapping modern semi-endogenous estimates into the gap specification places θ below $\bar{\theta}^{dest}$, in which case even correct beliefs would not amplify and the economy sits in the dormant region.

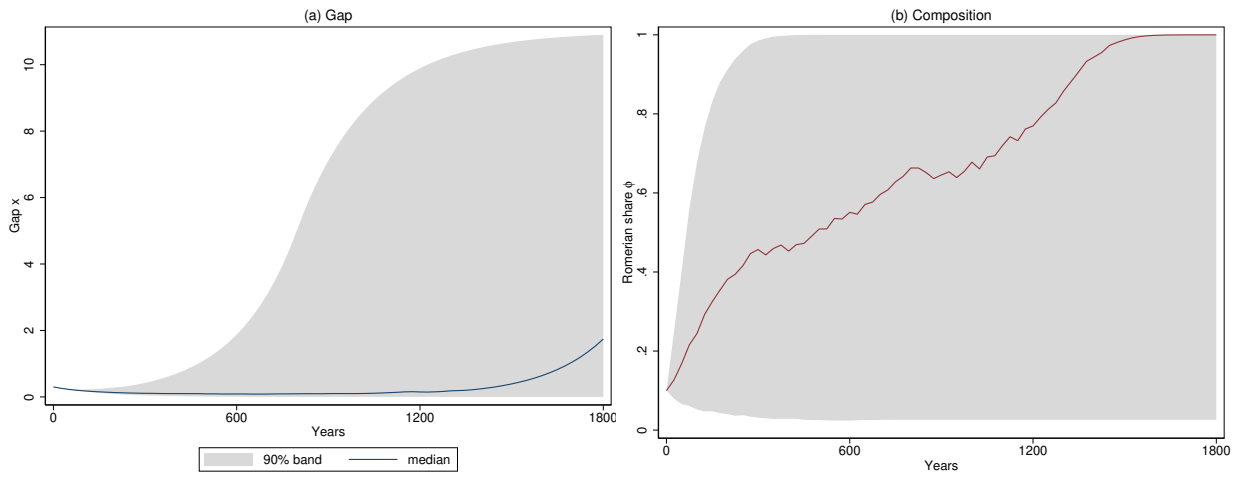


Figure 7. Monte Carlo Bands for the Benchmark Economy

Notes: Pointwise 5th–95th percentile bands and medians across 500 Monte Carlo replications of the productivity-shock sequence, drawn with a fixed seed, for the benchmark calibration with $\phi_0 = \pi_0 = 0.10$ and $x_0 = 0.30$ under the Romerian truth. A generation is 25 years.

5 Empirical Evidence: Beliefs and Fertility

The theoretical and quantitative analyses rest on a behavioral premise: beliefs about the long-run economic consequences of population shape fertility. This section provides supporting evidence from the General Social Survey, complemented by evidence from a historical natural experiment.

5.1 Data and Main Results

I use the GSS, a nationally representative survey of U.S. adults fielded in 2000 and 2010. The key belief variable asks respondents whether they agree that "The earth cannot continue to support population growth at its present rate," measured on a five-point scale.[2] I classify those who agree or strongly agree as holding Malthusian beliefs; respondents who do not endorse the statement serve as the comparison group and need not consist exclusively of Romerians in the model's sense. The sample contains 1,822 observations. Approximately 55 percent hold Malthusian beliefs. Mean ideal fertility is 2.50 children and mean actual fertility is 1.81. Appendix Table A.1 reports full summary statistics, and Appendix Figure A.1 displays the raw gradient between belief intensity and both fertility measures.

Table 3 reports OLS regressions of ideal and actual fertility on the Malthusian indicator. With baseline controls (age, age squared, sex, education, and year fixed effects), Malthusians report

[2]The original `POPGRWTH` coding runs from strong agreement to strong disagreement. The continuous measure used here reverses that ordering, so larger values indicate stronger Malthusian beliefs. Non-substantive responses are excluded.

0.27 fewer ideal children and have 0.29 fewer actual children, corresponding to approximately 11 and 16 percent of the respective sample means. Adding marital status, race, religion, political views, and income attenuates these coefficients to -0.20 and -0.24, but both remain significant at the one-percent level, preserving roughly three-quarters of the baseline association. That the actual-fertility coefficient is at least as large as the ideal-fertility coefficient indicates the association extends beyond stated preferences to realized behavior. Replacing the binary indicator with the continuous five-point scale yields comparable results (Appendix C).

Table 3. Malthusian Beliefs and Fertility

	Ideal Fertility		Actual Fertility	
	(1)	(2)	(3)	(4)
Malthusian	-0.268^{***}	-0.203^{***}	-0.292^{***}	-0.237^{***}
	(0.041)	(0.045)	(0.066)	(0.071)
Age	-0.014^{**}	-0.008	0.112^{***}	0.074^{***}
	(0.006)	(0.008)	(0.010)	(0.013)
Female	0.015	-0.006	0.091	-0.060
	(0.040)	(0.043)	(0.068)	(0.070)
Education	-0.047^{***}	-0.032^{***}	-0.144^{***}	-0.118^{***}
	(0.007)	(0.008)	(0.013)	(0.014)
Year FE	Yes	Yes	Yes	Yes
Additional controls	No	Yes	No	Yes
Observations	1,822	1,567	1,822	1,567
R^2	0.060	0.099	0.262	0.333

Notes: OLS with robust standard errors in parentheses. All models include age, age squared, sex, education, and year fixed effects; the reported age coefficient is the linear term. Additional controls in columns 2 and 4 are marital status, race, religion, religiosity, political views, and log real income. Sample size falls as covariates with missing values are included. Appendix Table A.2 traces the coefficient across intermediate control sets; Appendix C reports the continuous belief scale and other alternative specifications. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

How important is the belief variable relative to standard predictors? Figure 8 reports a Shapley–Owen R^2 decomposition that allocates explained variance fairly across seven predictor groups regardless of their entry order, with age, sex, and year fixed effects partialled out as base controls. For ideal fertility, Malthusian belief ranks third among the seven groups, accounting for roughly one-fifth of the explained variation above the base—behind race and education but ahead of religion, income, political views, and marital status. For actual fertility, marital status and education dominate, together explaining about 70 percent of the variation above the base; the Malthusian belief contributes a smaller share, comparable to that of income and larger than that of political views. Figure 9 provides a complementary comparison through standardized regression coefficients. For

ideal fertility, a one-standard-deviation increase in Malthusian belief is associated with a reduction similar in magnitude to the standardized effects of education, income, and religious attendance. That a single attitudinal variable explains a share of the variation comparable to broad socioeconomic predictors reinforces the evidence that population beliefs are nontrivially associated with fertility, independent of standard demographic controls.

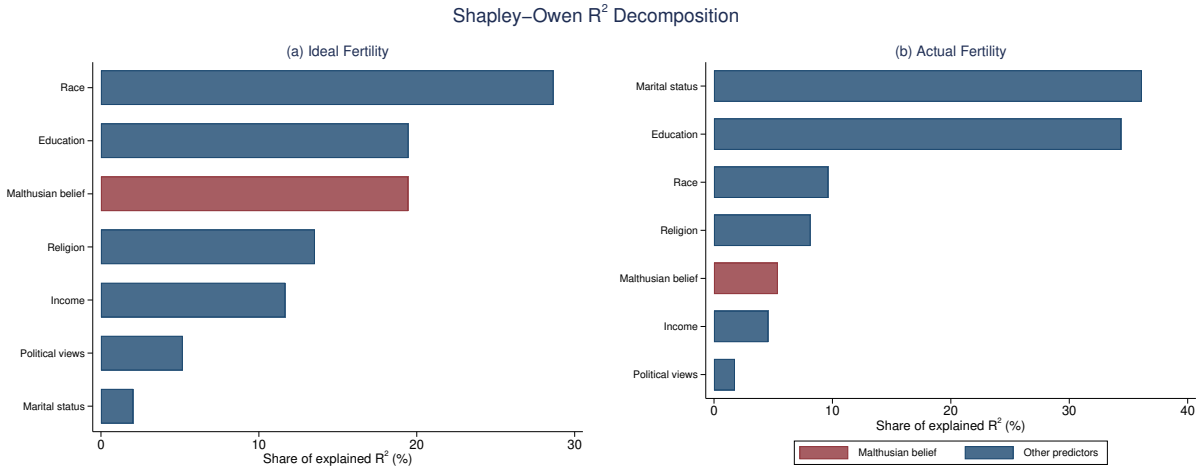


Figure 8. Shapley–Owen R^2 Decomposition: Relative Predictor Importance

Notes: Each bar shows a predictor group's share of the R^2 above base controls (age, age squared, sex, and year fixed effects). The Shapley–Owen method allocates the explained variance across the seven groups by averaging marginal contributions over all possible orderings. The total ΔR^2 attributable to the seven groups and the full model R^2 are reported on each panel. Sample restricted to observations with nonmissing values for all predictors ($N = 1{,}567$).

5.2 Robustness

The association is robust along several dimensions; Appendix C reports the complete results. The Malthusian coefficient attenuates gradually as controls are added sequentially, with no single covariate group accounting for a disproportionate share (Appendix Table A.2 and Figure A.3). Bias-adjusted bounds following Oster (2019) give -0.153 for ideal fertility and -0.161 for actual fertility, indicating that proportional selection on unobservables would need to be substantially stronger than selection on the included covariates to eliminate the association. Decomposing actual fertility into extensive and intensive margins attributes the relationship primarily to the decision to have any children (odds ratio 0.55; Appendix Table A.4). The coefficient is stable across the 2000 and 2010 survey waves (Appendix Table A.5), persists among respondents likely to have completed childbearing (Appendix Table A.8), appears across education groups (Appendix Table A.9), and is essentially unchanged when a measure of general environmental concern is included (Appendix Table A.6). Three patterns weigh against reverse causality (Appendix Table A.7): the ideal-fertility

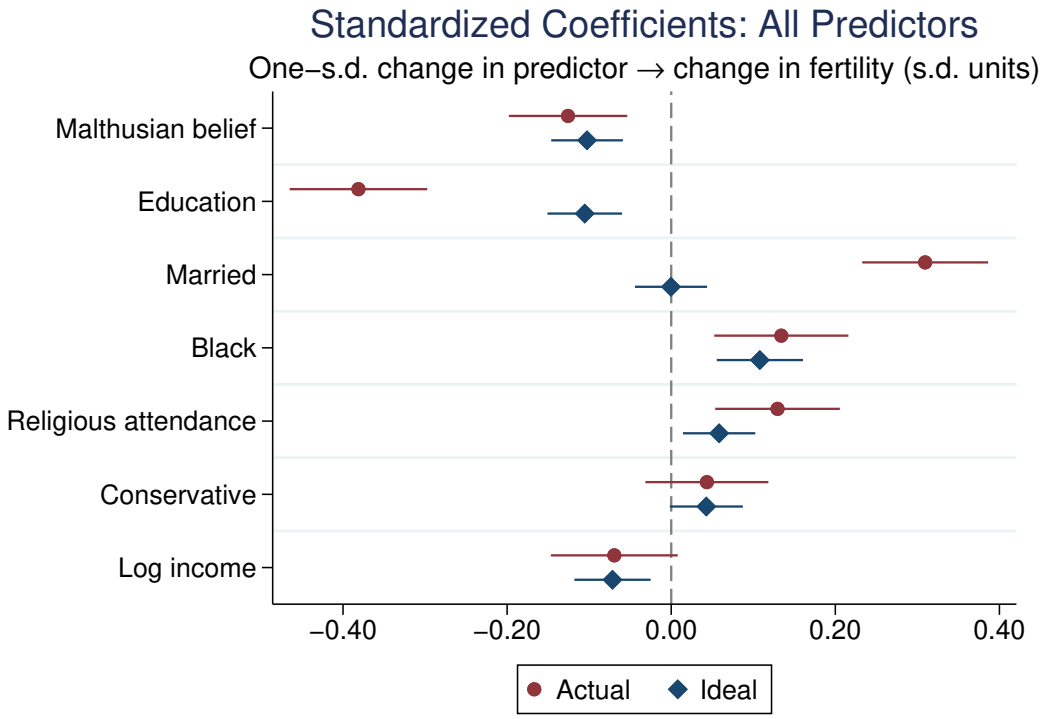


Figure 9. Standardized Coefficients: All Predictors

Notes: Each coefficient represents the change in fertility (in standard-deviation units) associated with a one-standard-deviation increase in the predictor. All variables are standardized to mean zero and unit variance. Religion enters through attendance frequency (continuous); denomination indicators are excluded because standardizing binary variables yields magnitudes that depend on the sample proportion rather than on effect size. All regressions include age, sex, and year fixed effects (also standardized). Bars show 95% confidence intervals based on robust standard errors. $N = 1{,}567$.

gradient is significant even among 18–34-year-olds with limited childbearing experience; the actual-fertility coefficient is zero among the young but large and significant among older respondents whose fertility histories more plausibly reflect accumulated decisions; and controlling for the respondent's number of siblings leaves the Malthusian coefficient essentially unchanged.

5.3 Cohort Variation and the State-Dependent Prediction

Proposition 1 predicts that the Malthusian–fertility gradient depends on the discounted expected population gap, X_t^{λ}. A necessary condition for this state-dependent mechanism is that the gradient varies across demographic trajectories – a constant gradient would instead be consistent with a time-invariant preference in which Malthusians always desire fewer children. This subsection provides suggstive evidence on the necessary condition by examining how the within-cohort Malthusian coefficient on ideal fertility covaries with the world population growth rate experienced by each birth cohort.

I assign each GSS respondent a birth year equal to the survey year minus age and group respondents into ten-year birth-cohort bins. The broader bins are necessary because splitting the 1,822-observation analysis sample into narrower intervals leaves too few respondents for reliable within-cohort estimation. Bins containing fewer than 100 observations are excluded, yielding six cohorts with midpoints ranging from the mid-1930s to the mid-1980s. For each cohort, I regress ideal fertility on the Malthusian indicator with a reduced set of controls: age, age squared, sex, education, marital status, and year fixed effects.[3]

To characterize the population-growth environment experienced by each cohort during its childbearing years, I compute the world population growth rate in the year the cohort midpoint reaches age 30, using annual data from the United Nations World Population Prospects.[4] Age 30 lies within the window during which childbearing decisions are typically made and fertility norms consolidated. World population growth peaked at approximately 2.1 percent per year in the mid-1960s and has since declined to roughly 1.1 percent by the mid-2010s. Cohorts born in the 1930s experienced this peak around age 30; those born in the 1980s reached age 30 during a period of substantially slower growth.

Figure 10 overlays the within-cohort Malthusian coefficients (left axis, navy points with 95 percent confidence intervals) and the world population growth rate at age 30 (right axis, maroon dashed line). Two patterns are visible. First, all six within-cohort coefficients are negative, confirming that the belief–fertility association is present in every decade-of-birth group rather than driven by a single generation. Second, the coefficient is most negative for the earliest cohort and becomes broadly less negative for later cohorts, tracking the secular decline in world population growth.

The broad comovement between the two series accords with the state-dependent prediction of Proposition 1. In the model, faster population growth relative to the adjusting reference makes the gap x_t more positive, raising the discounted expected gap X_t^{λ} and widening the fertility differential between types. Cohorts who reached childbearing age during the peak of world population growth plausibly perceived a larger positive gap, and the more negative Malthusian coefficient for these cohorts matches the model's implication that the belief–fertility gradient is strongest when the perceived gap is largest. As population growth slowed for later cohorts, the perceived gap narrowed, compressing the fertility differential toward zero. The pattern therefore satisfies the necessary condition for the state-dependent mechanism and is difficult to reconcile with a constant-preference alternative.

[3]The full control set from Table 3 is not employed in order to preserve degrees of freedom; the progressive-controls analysis in Table A.2 documents that the Malthusian coefficient changes only gradually as additional covariates are introduced, so the reduced specification provides a reliable estimate of the gradient within each bin.

[4]World population growth rates are computed as annual log differences from the UN World Population Prospects total-population series. The growth series is constructed on five-year cohort intervals

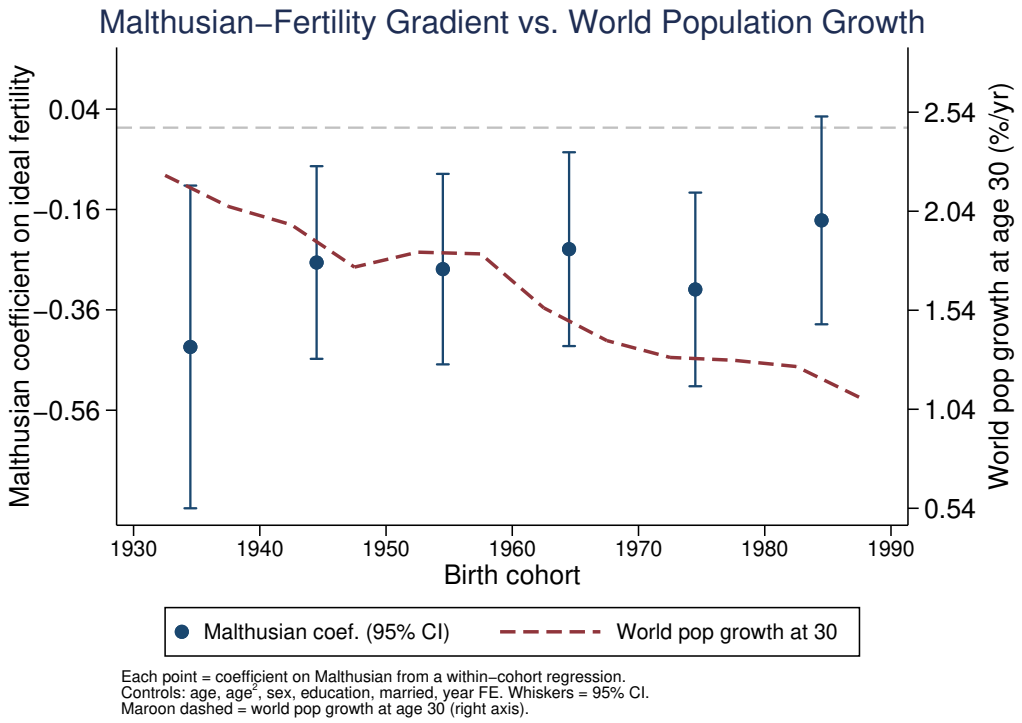


Figure 10. Malthusian–Fertility Gradient and World Population Growth by Birth Cohort

Notes: Each navy point is the OLS coefficient on the Malthusian indicator from a within-cohort regression of ideal fertility on belief, with whiskers showing 95% confidence intervals based on robust standard errors. Controls: age, age squared, sex, education, marital status, and year fixed effects. Cohorts are ten-year bins; bins with fewer than 100 observations are excluded. The maroon dashed line plots the world population growth rate (percent per year, right axis) in the year the cohort midpoint reaches age 30, computed from United Nations World Population Prospects data on five-year intervals with linear interpolation. The gray dashed line marks zero on the left axis.

5.4 Interpretation and Complementary Evidence

Proposition 1 makes the model's prediction conditional: near the reference, Romerians have higher fertility when households perceive a positive discounted gap forecast, $X_t^{\lambda} > 0$. The survey question presupposes ongoing growth, and the 2000 and 2010 waves were fielded against continued global population expansion, making a positive discounted expected gap plausible. The observed negative gradient is consistent with this region of the model. The survey does not directly observe respondents' demographic references or expected population paths, however, so the evidence should be read as consistent with the conditional prediction rather than as a direct test of the ranking reversal. The cohort-level evidence in Section 5.3 provides preliminary support for this variation by showing that the gradient covaries with the world population growth rate experienced during each birth cohort's childbearing years. A survey that jointly elicits scale-effect beliefs and expected population trajectories could test whether the gradient weakens near a perceived reference path and

reverses when the discounted expected gap becomes negative.

The estimated gaps also provide a useful scale for aggregate composition effects. Using the baseline actual-fertility coefficient as an empirical analogue of the group difference gives a gap of approximately 0.29 children. A ten-percentage-point increase in the Malthusian share corresponds to about 0.03 fewer children under fixed schedules—substantial relative to a 0.5-child fertility shortfall. These calculations hold the estimated group difference fixed and abstract from changes in forecasts, learning, transmission, and equilibrium fertility responses, which can amplify or offset the composition effect depending on the expected population path.

A recent paper by Robertson (2025) offers complementary evidence by showing that plausibly exogenously assigned Malthusian economic ideas change high-stakes behavior, though in bureaucratic relief rather than household fertility.[5] Together, the GSS associations, their descriptive aggregate magnitude, and the complementary evidence on the behavioral force of economic ideas provide suggestive support for the relevance of the belief channel. Establishing its causal contribution to aggregate fertility dynamics remains an empirical task.

Several limitations are worth noting. The evidence remains correlational; a fully convincing causal identification would require panel data or exogenous variation in beliefs. The belief measure captures concern about global population sustainability rather than the scale-effect parameter alone, and non-Malthusians need not hold the model's Romerian worldview. The cross-sectional design precludes tracking how belief changes affect subsequent fertility decisions. The GSS evidence should therefore be read as direct but observational evidence on the fertility margin.

6 Conclusion

This paper studies the implications of disagreement about population's economic consequences for fertility and growth. Parents who expect population growth to improve their descendants' prospects choose different fertility than parents who anticipate increasing population pressure. I embed this disagreement in a dynastic fertility model with cultural transmission and semi-endogenous productivity dynamics. The resulting fertility differential depends on the expected population path: Romerians have more children when the discounted expected gap is positive, and Malthusians have more when it is negative. Along an exact reference path, both types choose the same fertility, and the economy generates no evidence to distinguish their models.

The interaction between fertility and information gives rise to a dormant informational trap:

[5] Robertson (2025) studies the training of British Indian officials at Haileybury, where Malthus taught political economy. Malthus's death and his replacement by Richard Jones, a prominent critic of Malthusian doctrine, generated plausibly exogenous variation in the ideas to which successive cohorts of officials were exposed. Officials taught by Malthus later provided systematically less drought relief than officials taught by Jones, decades after leaving the classroom.

when fertility responses return the population gap toward the reference, observations become uninformative and the disagreement persists. A correct-belief comparison reveals the cost of inherited beliefs. An economy with correct Romerian beliefs amplifies a favorable demographic displacement, while an otherwise identical economy burdened by inherited pessimism contracts it back toward dormancy. In the calibrated illustration, the entry delay exceeds a millennium at the benchmark composition, and the initial Romerian share—rather than the underlying technology—determines whether the economy stagnates or grows. Wide dispersion in takeoff timing across simulated productivity histories highlights a role for cultural entrepreneurship in crossing the amplification threshold. General Social Survey evidence supports the behavioral premise: respondents with Malthusian beliefs report lower desired and actual fertility, with the association robust to extensive controls and economically meaningful in magnitude.

Two directions for future research follow naturally. The first concerns modern fertility decline. Fertility has fallen below replacement across much of the developed world, accompanied by widespread concern about planetary carrying capacity. Whether this pessimism reflects a genuine shift in the population–technology relationship or the cultural spread of beliefs that may not match the underlying fundamentals is an open question with direct policy implications. The model provides a framework for studying it: the state-dependent ranking implies that a larger Malthusian share lowers aggregate fertility while the discounted expected gap is positive but raises it once the gap turns negative, and the dormant trap identifies conditions under which inherited pessimism persists even if the truth is Romerian. A formal stationary analysis appears in Appendix A.10, and a calibrated illustration appears in Appendix D.4.

The second concerns population forward guidance. Standard family policies, e.g., subsidies, parental leave, childcare provision, have modest effects on fertility. The model suggests a complementary channel: credible government communication about the population–technology relationship could shift the social credence and thereby affect fertility through belief composition, much as central bank forward guidance shifts private expectations by announcing an assessment of fundamentals. The limitations are analogous—the government may not know the true scale-effect parameter, and credibility is constrained when experts themselves disagree—but the mechanism identifies a potential role for policy in coordinating beliefs rather than subsidizing births.

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A Additional Results and Proofs

This appendix states the technical conditions, supplies additional response formulas, and proves the results in the main text. Local equilibrium results are conditional on the specified smooth dormant branch. The Gaussian learning identities are exact.

Write

$$k = 1 - \rho, \qquad r = 1 - \mu, \qquad q_n = 1 - \varepsilon.$$

The exponent q_n is distinct from the contraction factor used below.

A.1 Feasibility and the Household Problem

The time cost of raising n children is

$$\psi(n) = 1 - \left[1 - (1-\rho)\frac{\gamma}{2}n^2\right]^{1/(1-\rho)}, \qquad 0 \le n < n_{\max}, \qquad n_{\max} = \sqrt{\frac{2}{(1-\rho)\gamma}}, \tag{A.1}$$

with $\gamma > 0$. The required time increases with fertility and approaches the entire endowment at the upper limit. This specification gives quadratic flow utility after removing the productivity scale:

$$f(n) \equiv u(1 - \psi(n)) = \frac{1}{1-\rho} - \frac{\gamma}{2}n^2, \tag{A.2}$$

which yields a simple marginal cost of fertility in utility units and a strictly concave private choice problem.

A child inherits the parent's worldview with probability λ; otherwise the child is Romerian with probability the updated credence π'. The child-type transition probabilities are

$$P_{jR}(\pi') = \lambda \mathbb{1}\{j = R\} + (1-\lambda)\pi', \qquad P_{jM}(\pi') = 1 - P_{jR}(\pi'). \tag{A.3}$$

Assumption 1 (Feasibility and finite value)**.** Fertility belongs to

$$\mathcal{N} = [\underline{n}, \overline{n}] \subset (0, n_{\max}), \qquad \underline{n} < 1 < \overline{n}.$$

The initial gap belongs to

$$\mathcal{X} = \left[\frac{\ln \underline{n}}{\mu}, \frac{\ln \overline{n}}{\mu}\right].$$

Define

$$\overline{\kappa} = \max_{\substack{j \in \{R,M\} \\ x \in \mathcal{X}}} \exp\left\{(1-\rho)\theta_j x + \frac{1}{2}(1-\rho)^2\sigma^2\right\}.$$

Assume

$$\beta \overline{n}^{1-\varepsilon} \overline{\kappa} < 1. \tag{A.4}$$

Fertility bounds are inactive in the local analysis.

Normalized expected child value is

$$C_j(z) = \mathbb{E}_j \left[\left(\frac{A'}{A} \right)^{1-\rho} \sum_{\ell \in \{R,M\}} P_{j\ell}(\pi') w_\ell(z') \right]. \tag{A.5}$$

Time costs, homogeneity, and finite value. Let

$$B(n) = 1 - (1-\rho)\frac{\gamma}{2} n^2.$$

The time-cost specification implies

$$1 - \psi(n) = B(n)^{1/(1-\rho)}, \qquad \psi'(n) = \gamma n B(n)^{\rho/(1-\rho)}.$$

Consequently,

$$(1-\psi(n))^{-\rho} \psi'(n) = \gamma n, \tag{A.6}$$

and

$$u(1-\psi(n)) = \frac{1}{1-\rho} - \frac{\gamma}{2} n^2,$$

which establishes (A.2).

For a given scale-invariant aggregate transition, the normalized Bellman equation is

$$w_j(z) = \max_{n \in \mathcal{N}} \left\{ f(n) + \beta n^{q_n} \mathbb{E}_j \left[e^{ks} \sum_\ell P_{j\ell}(\pi') w_\ell(z') \right] \right\}.$$

The state transition is independent of the productivity level, and

$$\mathbb{E}_j[e^{ks} \mid z] = \exp\left\{ k\theta_j x + \frac{1}{2} k^2 \sigma^2 \right\} \leq \overline{\kappa}.$$

Since transition probabilities sum to one, the Bellman operator is a contraction on bounded normalized values with modulus at most

$$\beta \overline{n}^{q_n} \overline{\kappa} < 1.$$

Scaling productivity scales each feasible consumption by the same factor and utility by that

factor to the power k, establishing (10). The gap interval is invariant because

$$x' = rx + \mu\left(\frac{\ln \bar{n}}{\mu}\right)$$

is a convex combination of points in $\mathcal{X}$.

Continuation value is positive: flow utility is positive on the feasible interval, and Bellman iteration from zero yields positive values. For a fixed aggregate transition, continuation value is independent of individual fertility. The objective's second derivative is

$$-\gamma - \beta\varepsilon(1-\varepsilon)n^{-1-\varepsilon}C_j < 0.$$

Thus the optimum is unique, and its interior first-order condition is (11). □

The illustrative cost calibration is

$$\gamma = \frac{2(1-0.85^{1-\rho})}{1-\rho}.$$

A.2 The Reference Household Problem

Define the reference utility-scale multiplier and the effective reference discount factor by

$$\kappa = \exp\left\{\frac{1}{2}(1-\rho)^2\sigma^2\right\}, \qquad \delta = \beta^\dagger\kappa \in (0,1), \tag{A.7}$$

where $\beta^\dagger$ is the replacement normalization defined in (A.9) below. The common interior reference fertility solves

$$\gamma(n^0)^{1+\varepsilon} = \frac{\beta(1-\varepsilon)\kappa f(n^0)}{1-\beta\kappa(n^0)^{1-\varepsilon}}. \tag{A.8}$$

The replacement normalization is

$$\beta^\dagger = \frac{\gamma}{\kappa[\gamma + (1-\varepsilon)f(1)]}. \tag{A.9}$$

Provided this choice is admissible and satisfies Assumption 1,

$$\delta = \frac{\gamma}{\gamma + (1-\varepsilon)f(1)} \in (0,1), \qquad w^0 = \frac{f(1)}{1-\delta}. \tag{A.10}$$

At $\rho = \varepsilon = 0.5$, $\sigma^2 = 0.04$, and the illustrative cost calibration,

$$\kappa \simeq 1.0050, \qquad \beta^\dagger \simeq 0.2517, \qquad \delta \simeq 0.2530.$$

Proof of Lemma 1. Under a permanently zero-gap forecast, both types predict the same productivity distribution. Every child-type transition matrix preserves a pair of common constant values. The normalized reference problem is

$$w = \max_{n \in \mathcal{N}} \{f(n) + \beta\kappa n^{1-\varepsilon} w\}.$$

Contraction gives a unique value, and strict concavity gives a unique choice. At that choice,

$$w = \frac{f(n^0)}{1 - \beta\kappa(n^0)^{1-\varepsilon}}.$$

Substitution into the fertility condition proves (A.8).

At replacement, the Bellman identity and first-order condition require

$$w^0 = \frac{f(1)}{1 - \beta\kappa}, \qquad \gamma = \beta(1-\varepsilon)\kappa w^0.$$

Solving gives (A.9). Strict concavity and contraction establish that the candidate solves the reference problem. Equation (A.10) follows. □

A.3 Local Conditions and the Fertility Ranking

Assumption 2 (Local differentiability)**.** Household responses are differentiable at the dormant path, and the discounted first-order continuation terms satisfy the terminal conditions stated below. For equilibrium calculations, values and policies are twice continuously differentiable near the relevant dormant states, differentiation under conditional expectations is valid, and the equilibrium branch satisfies

$$n_R(0, \phi, \pi) = n_M(0, \phi, \pi) = 1, \qquad w_R(0, \phi, \pi) = w_M(0, \phi, \pi) = w^0$$

throughout the relevant belief neighborhood.

Consider a deterministic first-order common forecast

$$x_t^e(h) = hz_t + o(h)$$

around a dormant path with $\phi = \pi = p$. Forecast perturbations and their remainders satisfy the conditions needed for the discounted derivatives below. In addition to X_t^λ from (12), define the discounted gap governing the average response,

$$X_t^1 = \sum_{k=0}^{\infty} \delta^k x_{t+k}^e. \tag{A.11}$$

Using $K, \bar{\theta}_p$ from (17), the individual responses are

$$n_{j,t} - 1 = K\left[\bar{\theta}_p X_t^1 + (\theta_j - \bar{\theta}_p) X_t^\lambda\right] + o(h). \tag{A.12}$$

Their difference is (13).

At the baseline, the child-type transition matrix is

$$P^0 = \begin{pmatrix} \lambda + (1-\lambda)p & (1-\lambda)(1-p) \\ (1-\lambda)p & \lambda + (1-\lambda)(1-p) \end{pmatrix}. \tag{A.13}$$

The row vector $(p, 1-p)$ is invariant under P^0. The difference between the rows is $\lambda(1, -1)$.

Proof of Proposition 1. Write

$$w_{j,t}(h) = w^0 + h v_{j,t} + o(h), \qquad n_{j,t}(h) = 1 + h u_{j,t} + o(h).$$

A derivative of the transition probabilities multiplies the common reference value and sums to zero. The envelope theorem removes the current optimal-fertility derivative from the value derivative. Hence

$$v_{j,t} = \delta\left[k\theta_j w^0 z_t + \sum_\ell P^0_{j\ell} v_{\ell,t+1}\right]. \tag{A.14}$$

Reference values are constant across belief states, so their first-order perturbations have no separate first-order value effect.

Set

$$\bar{v}_t = p v_{R,t} + (1-p) v_{M,t}, \qquad d_t = v_{R,t} - v_{M,t}.$$

The matrix properties give

$$\bar{v}_t = \delta[k\bar{\theta}_p w^0 z_t + \bar{v}_{t+1}]$$

and

$$d_t = \delta[k\Delta\theta w^0 z_t + \lambda d_{t+1}].$$

The terminal conditions are

$$\lim_{m\to\infty} \delta^m \bar{v}_{t+m} = 0, \qquad \lim_{m\to\infty} (\delta\lambda)^m d_{t+m} = 0.$$

Bounded forecast coefficients and bounded derivative values are sufficient. These conditions also hold for a geometric first-order equilibrium path when $\delta|a| < 1$.

Forward iteration yields

$$\begin{aligned} \bar{v}_t &= \delta k w^0 \bar{\theta}_p \sum_{m=0}^{\infty} \delta^m z_{t+m}, \\ d_t &= \delta k w^0 \Delta\theta \sum_{m=0}^{\infty} (\delta\lambda)^m z_{t+m}. \end{aligned} \tag{A.15}$$

Differentiating the fertility condition and using

$$\gamma = \beta^\dagger (1-\varepsilon)\kappa w^0$$

gives

$$u_{j,t} = \frac{k\theta_j w^0 z_t + \sum_\ell P^0_{j\ell} v_{\ell,t+1}}{(1+\varepsilon)w^0} = \frac{v_{j,t}}{\delta(1+\varepsilon)w^0}.$$

Since

$$v_{j,t} = \bar{v}_t + \frac{\theta_j - \bar{\theta}_p}{\Delta\theta} d_t,$$

substitution proves (A.12) and (13). A nonzero leading discounted-gap coefficient determines the local sign. □

Lemma 2 (Constant-fertility forecasts)**.** *If expected log aggregate fertility is constant at m^e, then*

$$X^\lambda_t = \frac{x_t + \dfrac{\delta\lambda}{1-\delta\lambda} m^e}{1-(1-\mu)\delta\lambda}. \tag{A.16}$$

At a stationary gap $x^* = m^e/\mu$,

$$X^{\lambda,*} = \frac{x^*}{1-\delta\lambda}.$$

Proof. The forecast gap satisfies

$$x^e_{t+m} = r^m x_t + \frac{m^e}{\mu}(1-r^m).$$

Summing with weights $(\delta\lambda)^m$ gives

$$X^\lambda_t = \frac{x_t}{1-\delta\lambda r} + \frac{m^e}{\mu}\left[\frac{1}{1-\delta\lambda} - \frac{1}{1-\delta\lambda r}\right],$$

which simplifies to (A.16). Summing a constant gap yields the stationary expression. □

A.4 Selection and Exact Learning Identities

Let $\mathcal{F}_t$ denote public information before observing s_t. The current gap, fertility policies, and signal information are $\mathcal{F}_t$-measurable.

The Gaussian likelihood of the growth signal under model j is

$$\ell_j(s \mid x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(s-\theta_j x)^2}{2\sigma^2}\right\}. \tag{A.17}$$

Bayes' rule gives

$$\pi_{t+1} = \frac{\pi_t \ell_R(s_t \mid x_t)}{\pi_t \ell_R(s_t \mid x_t) + (1-\pi_t)\ell_M(s_t \mid x_t)}. \tag{A.18}$$

The Romerian reproductive advantage, defined as the difference between the birth share and the adult share, is

$$B_t \equiv \phi_t^{birth} - \phi_t = \frac{\phi_t(1-\phi_t)(n_{R,t} - n_{M,t})}{\bar{n}_t}. \tag{A.19}$$

Define the log odds

$$\ell_t = \ln\left(\frac{\pi_t}{1-\pi_t}\right).$$

The exact log-odds update is

$$\ell_{t+1} - \ell_t = \frac{\Delta\theta x_t}{\sigma^2}\left[s_t - \frac{\theta_R + \theta_M}{2} x_t\right]. \tag{A.20}$$

Under the Romerian truth, its conditional mean is $\mathcal{I}_t$; under the Malthusian truth, its conditional mean is $-\mathcal{I}_t$. Its conditional variance is $2\mathcal{I}_t$ under either model.

Orient log odds toward the truth:

$$Z_t = \begin{cases} \ell_t, & \theta^* = \theta_R, \\ -\ell_t, & \theta^* = \theta_M. \end{cases} \tag{A.21}$$

Then

$$Z_{t+1} - Z_t = \mathcal{I}_t + \Delta M_{t+1}, \tag{A.22}$$

where

$$\mathbb{E}_t[\Delta M_{t+1}] = 0, \qquad \mathbb{E}_t[(\Delta M_{t+1})^2] = 2\mathcal{I}_t. \tag{A.23}$$

The innovation is conditionally Gaussian.

Since the logistic function is $1/4$-Lipschitz,

$$\mathbb{E}_t|\pi_{t+1} - \pi_t| \le \frac{1}{4}\left(\mathcal{I}_t + \sqrt{2\mathcal{I}_t}\right). \tag{A.24}$$

The local posterior drifts are

$$\begin{aligned}\mathbb{E}_t^{\theta_R}[\pi_{t+1}-\pi_t] &= 2\pi_t(1-\pi_t)^2\mathcal{I}_t + o(\mathcal{I}_t),\\ \mathbb{E}_t^{\theta_M}[\pi_{t+1}-\pi_t] &= -2\pi_t^2(1-\pi_t)\mathcal{I}_t + o(\mathcal{I}_t).\end{aligned} \tag{A.25}$$

Proof of Proposition 2. Subtracting the adult share from the birth share gives (A.19). Adding and subtracting the current social credence in the composition law gives (14). At an interior share, the selection term λB_t has the sign of the fertility differential, establishing the local claim.

For posterior drift, write $\pi = \Lambda(\ell)$. The first two derivatives are

$$\Lambda' = \pi(1-\pi), \qquad \Lambda'' = \pi(1-\pi)(1-2\pi).$$

The third derivative is bounded. A small-information Gaussian log-odds increment has third absolute moment $O(\mathcal{I}_t^{3/2})$. Under the Romerian truth, a second-order expansion gives

$$\mathbb{E}_t^{\theta_R}[\Delta\pi_t] = \pi(1-\pi)\mathcal{I}_t + \pi(1-\pi)(1-2\pi)\mathcal{I}_t + o(\mathcal{I}_t).$$

This simplifies to the first line of (A.25). Reversing the conditional mean gives the second line.

For the exact weak sign, let f_R, f_M be the conditional signal densities and fix an interior credence π. Under the Romerian truth,

$$\mathbb{E}_R[\pi'] = \pi\mathbb{E}_R\left[\frac{1}{\pi + (1-\pi)f_M/f_R}\right] \geq \frac{\pi}{\pi + (1-\pi)\mathbb{E}_R[f_M/f_R]} = \pi$$

by convexity and $\mathbb{E}_R[f_M/f_R] = 1$. Under the Malthusian truth,

$$u \mapsto \frac{\pi u}{1-\pi+\pi u}$$

is concave and $\mathbb{E}_M[f_R/f_M] = 1$, giving $\mathbb{E}_M[\pi'] \leq \pi$. Boundary credences remain unchanged. □

A.5 Local Equilibrium Coefficients

The coefficient $a(p)$ is defined in (15). Write

$$b(p) = a(p) - (1-\mu) = \left.\frac{\partial \ln \bar{n}(x,\phi,\pi)}{\partial x}\right|_{(0,p,p)} \tag{A.26}$$

for the aggregate log-fertility slope at the dormant state. Under the conditions of Proposition 3, individual policy slopes are

$$\left.\frac{\partial n_j}{\partial x}\right|_{(0,p,p)} = b(p) + \frac{K(\theta_j - \bar{\theta}_p)}{1 - \delta\lambda a(p)}. \tag{A.27}$$

The root of (18) connected to reference catch-up at zero average belief is

$$a(p) = \frac{1 + \delta(1-\mu) - \sqrt{[1-\delta(1-\mu)]^2 - 4\delta K\bar{\theta}_p}}{2\delta}, \tag{A.28}$$

provided the discriminant is positive.

If this branch reaches $a = 1$, the corresponding average-belief threshold is

$$\bar{\theta}^{dest} = \frac{\mu(1-\delta)}{K}. \tag{A.29}$$

At the illustrative preferences and the quantitative calibration's $\mu = 0.10$, the threshold is approximately 0.224.

Proof of Proposition 3. Evaluate derivatives at $(0, p, p)$, and write

$$v_j = \left.\frac{\partial w_j}{\partial x}\right|_{(0,p,p)}, \qquad u_j = \left.\frac{\partial n_j}{\partial x}\right|_{(0,p,p)}.$$

Values and policies are constant along the dormant manifold, so their belief-state derivatives vanish there. Derivatives of transition probabilities sum to zero when multiplied by the common reference value. The envelope equation is

$$v_j = \delta\left[k\theta_j w^0 + a\sum_\ell P^0_{j\ell} v_\ell\right]. \tag{A.30}$$

The fertility derivative is

$$u_j = \frac{k\theta_j w^0 + a\sum_\ell P^0_{j\ell} v_\ell}{(1+\varepsilon)w^0} = \frac{v_j}{\delta(1+\varepsilon)w^0}. \tag{A.31}$$

Averaging with weights $(p, 1-p)$ gives

$$\bar{v} = \delta[k\bar{\theta}_p w^0 + a\bar{v}], \qquad \bar{v} = \frac{\delta k\bar{\theta}_p w^0}{1 - \delta a}.$$

At reference fertility one,

$$b = pu_R + (1-p)u_M = \frac{K\bar{\theta}_p}{1-\delta a}.$$

Using $b = a - r$ proves (18).

Subtracting the envelope equations gives

$$v_R - v_M = \delta[k\Delta\theta w^0 + a\lambda(v_R - v_M)].$$

Hence

$$u_R - u_M = \frac{K\Delta\theta}{1-\delta\lambda a}.$$

Combining the average and difference proves (A.27) and (19).

The coefficient equation is

$$\delta a^2 - (1+\delta r)a + r + K\bar{\theta}_p = 0.$$

Its specified root is (A.28), with

$$\frac{\partial a}{\partial \bar{\theta}_p} = \frac{K}{\sqrt{(1-\delta r)^2 - 4\delta K\bar{\theta}_p}} > 0.$$

Substituting $a = 1$ gives (A.29). □

The two algebraic roots solve a local derivative-consistency equation. Establishing global equilibrium multiplicity would require the full equilibrium map and an equilibrium-selection analysis.

A.6 Information Accumulation and Population Composition

Proof of Theorem 1. Summing the truth-oriented log-odds update gives

$$Z_T = Z_0 + \mathcal{C}_T + M_T, \qquad \langle M \rangle_T = 2\mathcal{C}_T.$$

On the finite-information event, the martingale converges almost surely. For localization, stop when $\mathcal{C}_T$ first exceeds an integer bound. Information increments are uniformly bounded under Assumption 1, so each stopped martingale has bounded expected quadratic variation. Almost-sure convergence of these stopped martingales, followed by a union over the bounds, covers the finite-information event. The log odds then have a finite real limit, giving an interior posterior limit.

For infinite cumulative information, consider

$$W_T = \sum_{t=0}^{T-1} \frac{\Delta M_{t+1}}{1 + \mathcal{C}_{t+1}}.$$

The weights are predictable because $\mathcal{I}_t$ is known at time t. Moreover,

$$\begin{aligned} \langle W \rangle_\infty &= \sum_{t=0}^{\infty} \frac{2\mathcal{I}_t}{(1 + \mathcal{C}_{t+1})^2} \\ &\leq 2 \int_0^\infty \frac{du}{(1+u)^2} = 2. \end{aligned}$$

The inequality uses the decreasing integrand on each cumulative-information interval. Thus W_T converges almost surely. Kronecker's lemma yields

$$\frac{M_T}{1 + \mathcal{C}_T} \longrightarrow 0$$

on the infinite-information event. Consequently $Z_T \to +\infty$, and the posterior identifies the truth. □

Corollary 1 (From credence to population composition)**.** *Suppose the social credence identifies the truth.*

If $\lambda < 1$ and correct-type fertility is eventually at least as high as incorrect-type fertility, the adult share of the correct type converges to one.

If $\lambda = 1$, the correct type has a positive initial share, and its fertility ratio relative to the incorrect type is eventually bounded below by $1 + \epsilon$ for some $\epsilon > 0$, its adult share also converges to one.

Proof. Let q_t^* be the adult share of the correct type and p_t^* its social credence, with $p_t^* \to 1$.

If correct-type fertility is eventually at least as high as incorrect-type fertility, the correct birth share is at least q_t^*. For $\lambda < 1$,

$$1 - q_{t+1}^* \leq \lambda(1 - q_t^*) + (1 - \lambda)(1 - p_{t+1}^*).$$

Iteration of this stable inequality gives $q_t^* \to 1$.

With perfect transmission,

$$\frac{q_{t+1}^*}{1 - q_{t+1}^*} = \frac{q_t^*}{1 - q_t^*} \frac{n_{correct,t}}{n_{incorrect,t}}.$$

An eventually uniform fertility ratio above one makes these odds diverge whenever the correct type

is initially present with share below one. Positive lower fertility prevents extinction in finitely many periods. A correct share already equal to one remains there. □

A.7 Uniform Return and Dormancy

Assumption 3 (Uniform local return)**.** Fix $p \in (0,1)$ and $d > 0$ with

$$[p-d, p+d] \subset (0,1).$$

Assumption 2 holds on a neighborhood of

$$\{0\} \times [p-d, p+d]^2.$$

For some $q_0 \in [0,1)$,

$$\sup_{\phi,\pi \in [p-d,p+d]} |F_x(0,\phi,\pi)| \leq q_0. \tag{A.32}$$

For a smooth branch, a strict coefficient bound at $(0,p,p)$ extends to such a neighborhood after reducing d. Exact dormancy holds while its zero-gap belief trajectory remains on the maintained dormant branch. In particular, it holds globally if the branch is dormant at every belief state encountered along that trajectory.

The nearby result in Theorem 2 admits the following additional bounds. For any $q \in (q_0, 1)$, there exist $r_x > 0$, $C < \infty$, and an event E such that, for $|x_0| \leq r_x$,

$$\mathbb{P}^{\theta^*}(E) \geq 1 - \frac{C|x_0|}{d}. \tag{A.33}$$

On E,

$$|x_t| \leq q^t |x_0|, \qquad \sum_{t=0}^{\infty} \mathcal{I}_t \leq \frac{\Delta\theta^2 x_0^2}{2\sigma^2(1-q^2)}. \tag{A.34}$$

The limiting credence satisfies

$$\mathbb{E}^{\theta^*}\left[\mathbb{1}_E |\pi_\infty - p|\right] \leq C|x_0|, \tag{A.35}$$

and therefore

$$\mathbb{P}^{\theta^*}\left(E \cap \{|\pi_\infty - p| > \epsilon\}\right) \leq \frac{C|x_0|}{\epsilon} \qquad \text{for every } \epsilon > 0.$$

The variable π_∞ can be assigned any value outside E for the expectation in (A.35).

On E, adult composition converges to π_∞ if $\lambda < 1$. With $\lambda = 1$, it converges to a possibly different limit.

Proof of Theorem 2. The proof first establishes the exact path and then bounds exit from a stable neighborhood.

Exact dormancy. At a zero gap, the two likelihoods coincide for every observation, so the social credence is unchanged. Both policies equal one on the dormant branch, preserving the zero gap. The composition recursion is

$$\phi_{t+1} = \lambda\phi_t + (1-\lambda)\pi_0,$$

with solution

$$\phi_t = \pi_0 + \lambda^t(\phi_0 - \pi_0).$$

Uniform contraction before exit. Define

$$\tau = \inf\{t \geq 0 : |\phi_t - p| \geq d \text{ or } |\pi_t - p| \geq d\}.$$

Uniform differentiability and (A.32) imply that, for sufficiently small $r_x > 0$,

$$|F(x, \phi, \pi)| \leq q|x|$$

throughout the gap and belief neighborhood. For $t < \tau$,

$$|x_t| \leq q^t|x_0|, \qquad |x_{t+1}| \leq q^{t+1}|x_0|.$$

Smoothness and equal reference fertility give $|B_t| \leq C_B|x_t|$. Thus

$$\sum_{t<\tau} |B_t| \leq \frac{C_B|x_0|}{1-q}. \tag{A.36}$$

Posterior variation. Using (A.24) and predictability of $\mathbb{1}\{t < \tau\}$,

$$\begin{aligned}\mathbb{E}\left[\sum_{t<\tau} |\pi_{t+1} - \pi_t|\right] &\leq \frac{1}{4}\mathbb{E}\left[\sum_{t<\tau}(\mathcal{I}_t + \sqrt{2\mathcal{I}_t})\right] \\ &\leq C|x_0|.\end{aligned}$$

Define

$$V_\pi = \sum_{t<\tau} |\pi_{t+1} - \pi_t|.$$

This includes the increment that may cause exit, and $\mathbb{E}[V_\pi] \leq C|x_0|$.

Exit probability. Composition satisfies

$$\phi_{t+1} - p = \lambda(\phi_t - p) + (1-\lambda)(\pi_{t+1} - p) + \lambda B_t.$$

Starting from $\phi_0 = \pi_0 = p$, iteration gives, up to and including the first exit,

$$\sup_{t \le \tau} |\phi_t - p| \le V_\pi + \lambda \sum_{t<\tau} |B_t|, \qquad \sup_{t \le \tau} |\pi_t - p| \le V_\pi.$$

For infinite stopping times, suprema are taken over finite dates. Exit implies that the right-hand bound is at least d. Markov's inequality gives

$$\mathbb{P}(\tau < \infty) \le \frac{C|x_0|}{d}.$$

Set $E = \{\tau = \infty\}$.

Convergence on the no-exit event. On E, contraction holds at every date, proving (A.34). Theorem 1 gives an interior posterior limit. The inequality

$$\mathbb{1}_E |\pi_\infty - p| \le V_\pi$$

proves (A.35) and its probability implication.

Absolute selection is summable. For $\lambda < 1$, the stable recursion

$$\phi_{t+1} = \lambda\phi_t + (1-\lambda)\pi_{t+1} + \lambda B_t$$

converges to π_∞. For $\lambda = 1$, the increments equal B_t and are absolutely summable, so the share also converges.

Finally,

$$\ln \bar{n}_t = x_{t+1} - (1-\mu)x_t.$$

Geometric gap convergence implies

$$\sum_t |\ln \bar{n}_t| < \infty.$$

Log population therefore converges to a finite value, and population converges to a positive finite level. Moreover,

$$\frac{\ln A_T - \ln A_0}{T} = \frac{\theta^*}{T} \sum_{t=0}^{T-1} x_t + \frac{1}{T} \sum_{t=0}^{T-1} \eta_{t+1} \longrightarrow 0$$

almost surely on E, by summability and the strong law for the independent shocks. □

A.8 Demographic Cycles: Conditions and Limits

This subsection gives exact diagnostics for the four-phase illustration and proves the local oscillation result. A global equilibrium following the entire four-phase sequence would require these conditions to hold successively along its path.

Phase Diagnostics

The gap increases precisely when

$$x_{t+1} - x_t = \ln \bar{n}_t - \mu x_t > 0. \tag{A.37}$$

Population increases when $\bar{n}_t > 1$.

In a Malthusian world, define

$$L_t^M = -\mathbb{E}_t^{\theta_M}[\pi_{t+1} - \pi_t] \geq 0.$$

The expected adult Romerian-share change is

$$G_t = \lambda B_t + (1-\lambda)(\pi_t - \phi_t) - (1-\lambda)L_t^M. \tag{A.38}$$

Within the region where the local ranking applies, a positive discounted gap gives a positive leading selection term, and a negative discounted gap gives a negative one.

Phase I requires an increasing positive gap and an increasing adult Romerian share. In expected-direction terms, the conditions are

$$\ln \bar{n}_t > \mu x_t, \qquad G_t > 0, \qquad X_t^\lambda > 0.$$

A realized-share description uses the actual composition increment in place of G_t.

Phase II requires

$$\ln \bar{n}_t < \mu x_t$$

for the positive gap to close. An expected decline in the adult Romerian share additionally requires $G_t < 0$. Their timing depends on aggregate fertility and the full composition law.

Crossing below the reference from a positive gap requires

$$\ln \bar{n}_t < -(1-\mu)x_t. \tag{A.39}$$

The ranking reversal in Phase III also requires $X_t^\lambda < 0$. Below the reference, the gap recovers when

$$\ln \bar{n}_t > \mu x_t.$$

Population recovery itself requires $\bar{n}_t > 1$.

Phase IV's convergence follows under the uniform-return conditions. A uniformly negative derivative supplies repeated overshooting. The local order comparison between selection and information leaves the global crossover and turning dates dependent on the equilibrium path.

Damped Oscillations

A precise local result follows when the stable gap response is negative throughout the relevant belief neighborhood.

Corollary 2 (Damped demographic cycles)**.** *Maintain the conditions of the nearby result in Theorem 2. Suppose that, for some* $0 < \alpha_0 \leq q_0 < 1$*,*

$$-q_0 \leq F_x(0, \phi, \pi) \leq -\alpha_0 \qquad \textit{for all } (\phi, \pi) \in [p-d, p+d]^2.$$

For sufficiently small $x_0 \neq 0$*, there is an event of probability at least* $1 - C|x_0|/d$ *on which*

$$\operatorname{sgn}(x_{t+1}) = -\operatorname{sgn}(x_t), \qquad |x_t| \leq q^t |x_0|$$

for some $q < 1$*. On that event, cumulative information is finite and the posterior limit is interior.*

Each generation's fertility response takes the gap across the reference and leaves a smaller displacement. Population growth also alternates around replacement along these paths. The corollary conditions on the smooth equilibrium branch and a uniformly negative response. The associated interval of average scale-effect beliefs is given in (A.41).

Repeated demographic reversals can therefore coexist with persistent disagreement. The squared gaps remain summable when their amplitudes decline geometrically, placing the path in the finite-information part of Theorem 1.

The additional condition in Corollary 2 is

$$-q_0 \leq F_x(0, \phi, \pi) \leq -\alpha_0 < 0 \qquad \text{for all } \phi, \pi \in [p-d, p+d], \tag{A.40}$$

where $0 < \alpha_0 \leq q_0 < 1$.

Proof of Corollary 2. Uniform continuity of the derivative gives constants $0 < \alpha < q < 1$ and a

sufficiently small gap neighborhood on which

$$-q \le F_x(x, \phi, \pi) \le -\alpha.$$

Since $F(0, \phi, \pi) = 0$, the mean value theorem implies, for every nonzero gap in the neighborhood,

$$\operatorname{sgn} F(x, \phi, \pi) = -\operatorname{sgn} x, \qquad \alpha|x| \le |F(x, \phi, \pi)| \le q|x|.$$

Apply the stopped-neighborhood construction of Theorem 2. On its no-exit event,

$$\operatorname{sgn} x_t = (-1)^t \operatorname{sgn} x_0, \qquad \alpha^t|x_0| \le |x_t| \le q^t|x_0|.$$

The event has probability at least $1 - C|x_0|/d$. Finite information and an interior posterior limit follow from the same theorem.

Because successive gaps have opposite signs,

$$\ln \bar{n}_t = x_{t+1} - (1 - \mu)x_t$$

has the sign of x_{t+1}. Population growth therefore alternates in sign as well. □

On the branch (A.28), the condition $-1 < a(p) < 0$ is equivalent to

$$-\frac{(2 - \mu)(1 + \delta)}{K} < \bar{\theta}_p < -\frac{1 - \mu}{K}. \tag{A.41}$$

This follows by evaluating

$$(a - r)(1 - \delta a) = K\bar{\theta}_p$$

at $a = -1$ and $a = 0$. Its left-hand side is strictly increasing over that interval.

For an interior belief state on a smooth branch, a coefficient strictly inside this interval extends to a sufficiently small neighborhood satisfying (A.40). The household boundedness restrictions and the maintained equilibrium-branch qualifications continue to apply.

Mean-Signal Periodic Orbits

A deterministic mean-signal comparison makes the cumulative nature of learning particularly transparent.

Proposition 5 (Periodic orbits under mean signals)**.** *Retain the equilibrium policy rules and replace each growth observation by*

$$s_t = \theta^* x_t.$$

If the policies equal replacement whenever $x = 0$, the resulting deterministic system has no nonconstant periodic orbit with an interior social credence.

Every nonzero gap then adds evidence in favor of the truth. An orbit returning to the same interior credence would have to contain only uninformative states, which implies a zero gap throughout. The comparison evaluates the stochastic model's policies along a mean-signal sequence; households retain the original signal model in forming those policies.

The full learning process can support demographic overshoots while retaining evidence from previous phases. Recurrent fixed-amplitude disagreement cycles would require additional forces that renew uncertainty or reduce the weight on past evidence, such as changing fundamentals, forgetting, or outside cultural shocks.

Proof of Proposition 5. Under mean signals,

$$Z_{t+1} - Z_t = \mathcal{I}_t \geq 0.$$

A periodic orbit of period m with an interior credence would require

$$0 = Z_{t+m} - Z_t = \sum_{s=t}^{t+m-1} \mathcal{I}_s.$$

Every term must be zero, implying a zero gap at every date on the orbit.

The maintained policy property gives replacement fertility. The credence is constant, and composition satisfies

$$\phi_{t+1} = \lambda\phi_t + (1-\lambda)\pi.$$

For $\lambda < 1$, only the fixed point of this contraction is periodic. For $\lambda = 1$, the share is constant. Hence every such periodic orbit is constant. □

A.9 The Correct-Belief Entry Comparison

Using the entry times in (22), define excess delayed-entry probability by

$$\mathcal{R}_h(T) = \mathbb{P}(T_h^H > T) - \mathbb{P}(T_h^C > T) \tag{A.42}$$

and the difference in truncated expected entry times by

$$\mathcal{D}_h(T) = \mathbb{E}[\min\{T_h^H, T\}] - \mathbb{E}[\min\{T_h^C, T\}]. \tag{A.43}$$

Under Proposition 4, for every $T \geq T_h^C$,

$$\mathcal{R}_h(T) \geq 1 - \frac{Cz}{d}.$$

Proof of Proposition 4. In the correct-belief economy, composition and credence remain at one. Scale invariance removes productivity-level shocks from normalized fertility, so its gap follows the deterministic map F_C.

For any $a_- \in (1, a_C)$, differentiability gives a sufficiently small $h > 0$ such that

$$F_C(x) \geq a_- x \qquad \text{for } 0 < x < h.$$

Iteration until entry yields

$$T_h^C \leq \left\lceil \frac{\ln(h/z)}{\ln a_-} \right\rceil. \tag{A.44}$$

Reduce h so that it also lies in the inherited economy's local neighborhood. On the no-exit event from Theorem 2,

$$|x_t^H| \leq q^t z < h$$

for every date. Thus $T_h^H = \infty$ on an event with probability at least $1 - Cz/d$.

For $T \geq T_h^C$, the correct-belief economy's delayed-entry probability is zero. The same no-exit event therefore gives the bound on $\mathcal{R}_h(T)$. □

A.10 The Fixed-Belief Stationary Comparison

The stationary comparison holds $\phi = \pi = p$ externally fixed and uses the constant child-type transition matrix $P(p)$ implied by (A.3). Bayesian updating and the population-composition law are suspended, while aggregate fertility and the stationary gap remain mutually consistent.

The common preference or cost perturbation is indexed by $\nu = \ln n^0$, so zero-gap household fertility is $n^0(\nu) = e^\nu$. All local response coefficients are evaluated at $\nu = 0$. Changes in those coefficients multiply the small gap and therefore enter beyond first order.

Proposition 6 (Stationary fertility adjustment)**.** *Consider the fixed-belief comparison above. Suppose household choices are interior and jointly differentiable in the gap and the common preference or cost perturbation. If $D(p) = \mu - K\bar{\theta}_p/(1-\delta) > 0$, a local stationary branch satisfies*

$$x^*(p,\nu) = \frac{\nu}{D(p)} + o(|\nu|), \qquad \ln \bar{n}^*(p,\nu) = \frac{\mu\nu}{D(p)} + o(|\nu|). \tag{A.45}$$

Its fertility differential is

$$n_R^* - n_M^* = \frac{K\Delta\theta}{1-\delta\lambda}\frac{\nu}{D(p)} + o(|\nu|). \tag{A.46}$$

For $\nu < 0$, the stationary gap is negative and Malthusians have higher fertility to first order. Across fixed shares with $D(p) > 0$, lowering p increases $D(p)$, reduces the first-order fertility decline, and reduces the magnitude of the gap and its signal information.

A downward shift in reference fertility creates a negative stationary gap. Malthusian beliefs then raise fertility relative to Romerian beliefs, offsetting part of the initial shift. Since this response also brings the gap closer to zero, it reduces the information content of each growth observation.

In addition to (A.45) and (A.46), signal information at the stationary gap satisfies

$$\mathcal{I}^* = \frac{\Delta\theta^2}{2\sigma^2}\frac{\nu^2}{D(p)^2} + o(\nu^2). \tag{A.47}$$

Proof of Proposition 6. For an imposed stationary gap, the aggregate derivative in (A.12) is $K\bar{\theta}_p/(1-\delta)$. At an imposed zero gap, both types choose e^ν, giving derivative one for log aggregate fertility with respect to ν. Joint differentiability implies

$$\ln \bar{n} = \nu + \frac{K\bar{\theta}_p}{1-\delta}x + o(|\nu| + |x|).$$

Stationarity requires $\ln \bar{n} = \mu x$, so

$$0 = D(p)x - \nu + o(|\nu| + |x|).$$

Since $D(p) > 0$, the implicit-function theorem gives (A.45).

The fertility differential is zero at an imposed zero gap for every small common perturbation. Its direct derivative with respect to ν is therefore zero. Its stationary gap derivative is

$$\frac{K\Delta\theta}{1-\delta\lambda}.$$

Substituting the stationary gap gives (A.46). Substitution into (7) gives (A.47).

Finally,

$$D'(p) = -\frac{K\Delta\theta}{1-\delta} < 0.$$

For $\nu < 0$, a lower Romerian share increases $D(p)$, makes the first-order fertility decline less negative, and reduces the magnitude of the gap and its information. □

A.11 The Learning-Time Bound

For the truth-oriented log odds Z_t in (A.21) and a target $d_Z > 0$, define

$$T_{d_Z} = \inf\{t \geq 0 : Z_t - Z_0 \geq d_Z\}.$$

Proposition 7 (An information-based learning-time bound)**.** *Suppose* $\pi_0 \in (0,1)$*,* $\nu \neq 0$*, and*

$$|x_t| \leq c_x|\nu| \qquad \textit{for } t < T_{d_Z},$$

where $c_x > 0$ *is independent of* ν*. If learning uses only the baseline productivity signals, then*

$$\mathbb{E}[T_{d_Z}] \geq \frac{2\sigma^2 d_Z}{\Delta\theta^2 c_x^2 \nu^2}. \tag{A.48}$$

The inequality also applies when the expected hitting time is infinite.

Proof. Until the stopping time,

$$\mathcal{I}_t \leq \overline{I}_\nu = \frac{\Delta\theta^2 c_x^2 \nu^2}{2\sigma^2}.$$

The result is immediate for infinite expected hitting time. Otherwise,

$$\mathbb{E}[\langle M\rangle_{T_{d_Z}}] = 2\mathbb{E}\left[\sum_{t<T_{d_Z}} \mathcal{I}_t\right] \leq 2\overline{I}_\nu \mathbb{E}[T_{d_Z}] < \infty.$$

The martingales stopped at $T_{d_Z} \wedge n$ are bounded in L^2. Optional stopping and L^2 convergence imply

$$\mathbb{E}[M_{T_{d_Z}}] = 0.$$

At the target,

$$Z_{T_{d_Z}} - Z_0 = \sum_{t<T_{d_Z}} \mathcal{I}_t + M_{T_{d_Z}} \geq d_Z.$$

Taking expectations gives

$$d_Z \leq \mathbb{E}\left[\sum_{t<T_{d_Z}} \mathcal{I}_t\right] \leq \overline{I}_\nu \mathbb{E}[T_{d_Z}],$$

which proves (A.48). □

B Interpretation and Extensions

B.1 The Population-Gap Specification

The reduced-form mechanism captures a temporary growth effect from a one-time population-level change and a persistent contribution from sustained population growth:

$$x_{t+1} = (1-\mu)x_t + \ln \bar{n}_t, \qquad \mathbb{E}_t[\Delta \ln A_{t+1}] = \theta^* x_t.$$

An exact derivation from an idea-production equation would generally impose additional restrictions. A gap defined using population and realized knowledge can transmit knowledge shocks into the gap and link its adjustment coefficient to technological parameters. The baseline permits a common positive catch-up rate across the rival scale-effect models and keeps productivity shocks out of the gap equation.

The reference index can fall when the gap is negative. Its interpretation is therefore an adjusting reference population. Convergence of the gap can occur at a population level different from the initial level.

B.2 The Zero-Intercept Baseline

A known common intercept can be added to productivity growth:

$$\Delta \ln A_{t+1} = \bar{g} + \theta^* x_t + \eta_{t+1}.$$

It cancels from the difference between rival signal means, leaving signal information unchanged. The reference utility-scale multiplier becomes

$$\kappa(\bar{g}) = \exp\left\{(1-\rho)\bar{g} + \frac{1}{2}(1-\rho)^2\sigma^2\right\}.$$

If the household discount factor is recalibrated to preserve replacement fertility at the reference,

$$\delta = \frac{\gamma}{\gamma + (1-\varepsilon)f(1)}$$

remains unchanged. The local response formulas expressed in terms of δ continue to apply. Long-run average log growth along a dormant path would equal $\bar{g}$. Holding the household discount factor fixed while changing the intercept would generally change reference fertility.

Positive productivity-shock variance distinguishes arithmetic expected growth from expected log growth. The zero-intercept normalization and the dormancy conclusions concern log growth.

B.3 The Role of Preferences

CRRA utility and time costs produce exact homogeneity, isolating the expected-growth channel from a direct productivity-level fertility effect. The restriction $\rho \in (0, 1)$ makes the specified flow utility positive, while diminishing dynastic weight and quadratic normalized costs give a strictly concave private problem.

The utility normalization is consequential with endogenous fertility. Adding a constant to flow utility can change the value assigned to additional descendants. The analysis maintains the stated

normalization throughout.

Common preferences ensure equal fertility when growth forecasts coincide. Type-specific preferences for children, child-rearing costs, or quantity–quality choices could preserve fertility differences at the reference. Such extensions would change the exact behavioral dormancy result.

Recursive altruism allows children to adopt a different worldview and evaluate their continuation accordingly. The association between the initial parent's type and descendants' types decays with parental transmission. This produces the discount factor $\delta\lambda$ in the fertility differential and δ in the average response.

B.4 Nonlinear Growth Functions

Consider the extension

$$\Delta \ln A_{t+1} = g(x_t; \theta^*) + \eta_{t+1}.$$

With a common Gaussian variance, signal information is

$$\mathcal{I}(x) = \frac{[g(x; \theta_R) - g(x; \theta_M)]^2}{2\sigma^2}.$$

The truth-oriented log-odds decomposition and cumulative-information criterion retain their form under suitable boundedness or localization conditions.

If the two growth functions are twice continuously differentiable near zero and satisfy

$$g(0; \theta_R) = g(0; \theta_M) = 0,$$

the local household formulas replace θ_j with

$$\vartheta_j = \frac{\partial g(0; \theta_j)}{\partial x}.$$

Opposite derivative signs preserve the local Romerian–Malthusian interpretation.

A global fertility ranking depends on continuation values along endogenous future paths. Extending the local ranking globally therefore requires restrictions on the full continuation problem in addition to a common crossing of current growth functions.

B.5 Conditions Affecting Dormancy

Differences at the reference. If the rival signal means differ at the reference, that state remains informative. A positive lower bound on information along the realized path gives posterior consistency. The cumulative-information criterion governs cases in which informativeness varies over

time.

Regularity near the crossing. Equality of the signal distributions at a single state gives exact informational silence there. Nearby dormancy additionally requires local bounds on the rate at which information and fertility differences vanish. The neighborhood theorem uses these regularity conditions to control the entire adjustment path.

Strong scale effects. With no reference catch-up, replacement fertility leaves the gap unchanged. An endogenous negative fertility response can still provide local return. This extension would require a separate invariant-state and finite-value argument.

State-dependent noise. With state-dependent variance, information depends on the squared difference in signal means relative to that variance. A variance that vanishes as fast as the squared gap can preserve information per observation near the reference. The baseline uses strictly positive constant variance.

Independent demographic shocks. Persistent disturbances entering the gap equation can make the sum of squared gaps diverge and restore learning. They can also interrupt damping by repeatedly displacing the economy from the reference. Their implications for exact dormancy and recurrent dynamics require an augmented state process.

Outside evidence. Regional comparisons, research, and other signals can identify scale effects without a domestic demographic departure. Infinite cumulative identifying information from such sources yields accurate social credence even if the domestic gap remains near zero.

Forgetting and changing fundamentals. Fixed-parameter Bayesian updating retains accumulated evidence. Forgetting, recency weighting, structural change, or outside cultural shocks can renew disagreement. These forces could support recurrent belief-driven episodes and would change the restriction on exact interior periodic orbits.

B.6 Static Responses and Equilibrium Dynamics

Equation (A.12) gives household responses to an imposed demographic forecast. Equation (18) adds consistency with the first-order aggregate continuation. Their derivatives correspond to different comparative-static and equilibrium objects.

For a scalar fixed-point problem $n = \Gamma(n)$, the condition $\Gamma'(n^*) \neq 1$ gives local uniqueness through the implicit-function theorem. Stability of iterative best-response adjustment instead

requires $|\Gamma'(n^*)| < 1$. Global multiplicity requires analysis of the full map and its boundary behavior.

The same distinction applies to the cycle illustration. The phase diagnostics describe the movements required for the expansion–correction sequence. Corollary 2 establishes alternating local contraction under its uniform negative-response condition. A global equilibrium characterization would determine which larger sequences the economy follows and how it enters the local region.

C Supporting Empirical Results

This appendix reports supplementary specifications and robustness checks for the empirical analysis in Section 5. All regressions use robust standard errors.

Table A.1. Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Ideal number of children	1,822	2.50	0.87	0	7
Actual number of children	1,822	1.81	1.64	0	8
Malthusian belief (binary)	1,822	0.55	0.50	0	1
Malthusian belief (1–5 scale)	1,822	3.44	1.03	1	5
Age	1,822	46.8	17.3	18	89
Years of education	1,822	13.3	2.96	0	20
Female	1,822	0.56	0.50	0	1

Notes: Data from GSS, 2000 and 2010 waves. Malthusian belief (binary) equals one if the respondent agrees or strongly agrees that Earth cannot continue to support population growth. Higher values of the continuous measure indicate stronger agreement.

Summary statistics and descriptive patterns.

Progressive controls. Table A.2 traces the Malthusian coefficient as the controls from Table 3 are added one group at a time: marital status, race, religion and religiosity, political views, and log real income. Columns 1 and 6 of each panel correspond to the baseline and full-controls specifications in the main table. The coefficient for ideal fertility declines from -0.268 to -0.203 and for actual fertility from -0.292 to -0.237, with all twelve intermediate estimates significant at the one-percent level.

Oster bounds. I follow Oster (2019) and compute

$$\beta^* = \tilde{\beta} - (\dot{\beta} - \tilde{\beta})\frac{R_{\max} - \tilde{R}^2}{\tilde{R}^2 - \dot{R}^2},$$

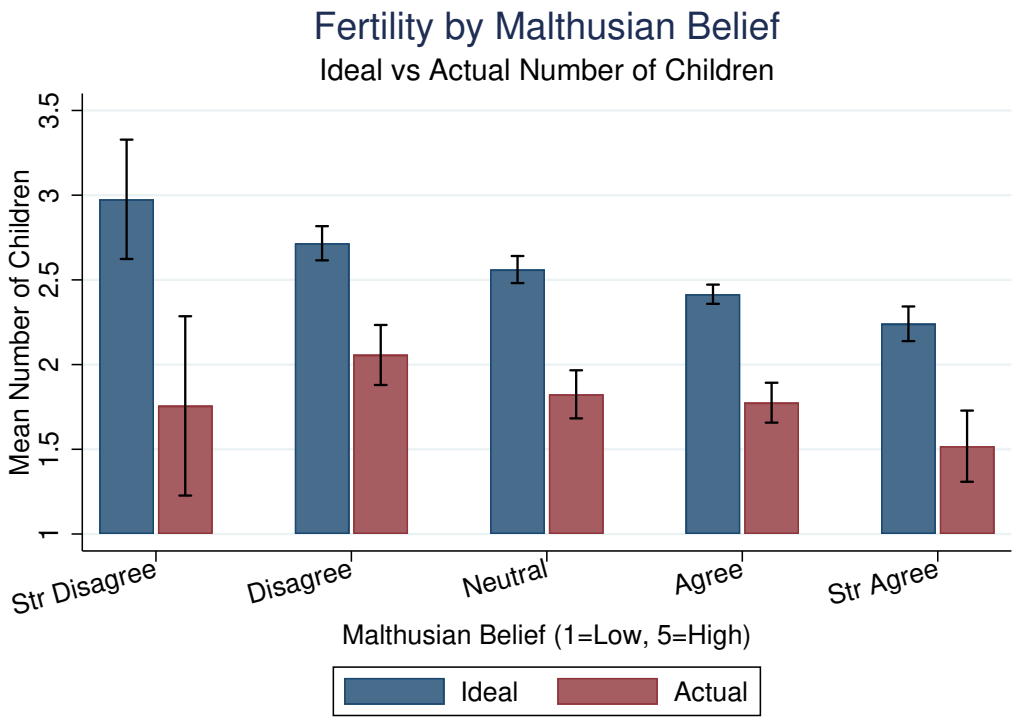


Figure A.1. Fertility by Malthusian Belief Intensity

Notes: Data from GSS, 2000 and 2010 waves ($N = 1{,}822$). Bars show mean ideal and actual numbers of children for each level of the Malthusian belief scale. Error bars indicate 95% confidence intervals.

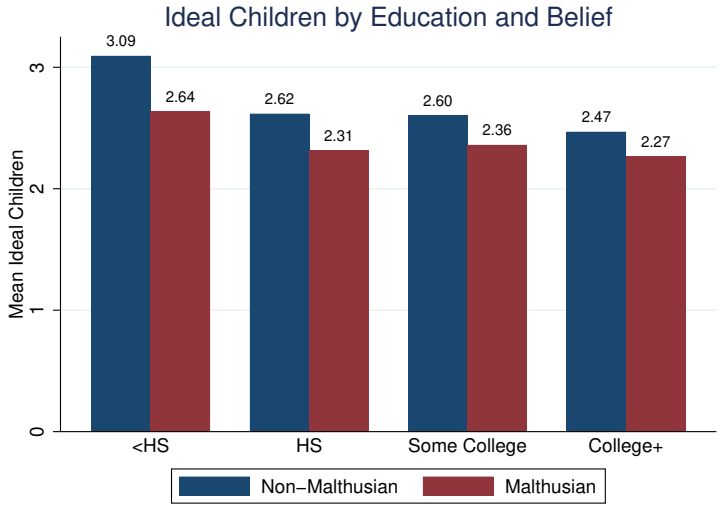


Figure A.2. Ideal Number of Children by Education and Malthusian Belief

Notes: Bars show mean ideal numbers of children for non-Malthusians and Malthusians within each education category.

Table A.2. Progressive Controls: Malthusian Coefficient

	(1) Baseline	(2) +Marital	(3) +Race	(4) +Religion	(5) +PolViews	(6) +Income
Panel A: Ideal Fertility						
Malthusian	-0.268^{***}	-0.272^{***}	-0.240^{***}	-0.220^{***}	-0.210^{***}	-0.203^{***}
	(0.041)	(0.041)	(0.041)	(0.042)	(0.042)	(0.045)
R^2	0.060	0.064	0.086	0.096	0.093	0.099
Panel B: Actual Fertility						
Malthusian	-0.292^{***}	-0.273^{***}	-0.224^{***}	-0.186^{***}	-0.207^{***}	-0.237^{***}
	(0.066)	(0.065)	(0.065)	(0.066)	(0.068)	(0.071)
R^2	0.262	0.311	0.327	0.336	0.338	0.333
N	1,822	1,822	1,822	1,809	1,740	1,567

Notes: Each column adds the indicated controls cumulatively to the baseline specification of Table 3; columns 1 and 6 correspond to the main table's baseline and full-controls estimates. Robust standard errors in parentheses. N declines as covariates with missing values are included. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

where $\dot{\beta}$ and $\dot{R}^2$ are from the baseline (short) regression and $\tilde{\beta}$ and $\tilde{R}^2$ are from the full-controls (long) regression. Setting $R_{\max} = \min\{1,\ 1.3 \times \tilde{R}^2\}$ and proportional selection $\delta = 1$:

	$\dot{\beta}$	$\tilde{\beta}$	$R_{\max}$	β^*
Ideal fertility	-0.268	-0.203	0.129	-0.153
Actual fertility	-0.292	-0.237	0.433	-0.161

Both bias-adjusted estimates are negative and exceed half the baseline coefficient in absolute value. Proportional selection on unobservables equal to that on observables is therefore insufficient to eliminate the association.

Count data and alternative classifications. Table A.3 presents alternative specifications. Poisson and negative binomial models yield significant Malthusian coefficients for both outcomes. Excluding neutral respondents strengthens the binary-indicator estimate to -0.357 for ideal and -0.342 for actual fertility. The continuous belief scale with full controls gives -0.127 and -0.105, respectively.

Extensive versus intensive margin. Table A.4 decomposes actual fertility into the probability of having any children (extensive margin) and the count conditional on parenthood (intensive margin). The extensive-margin odds ratio is 0.55 under baseline controls, corresponding to an

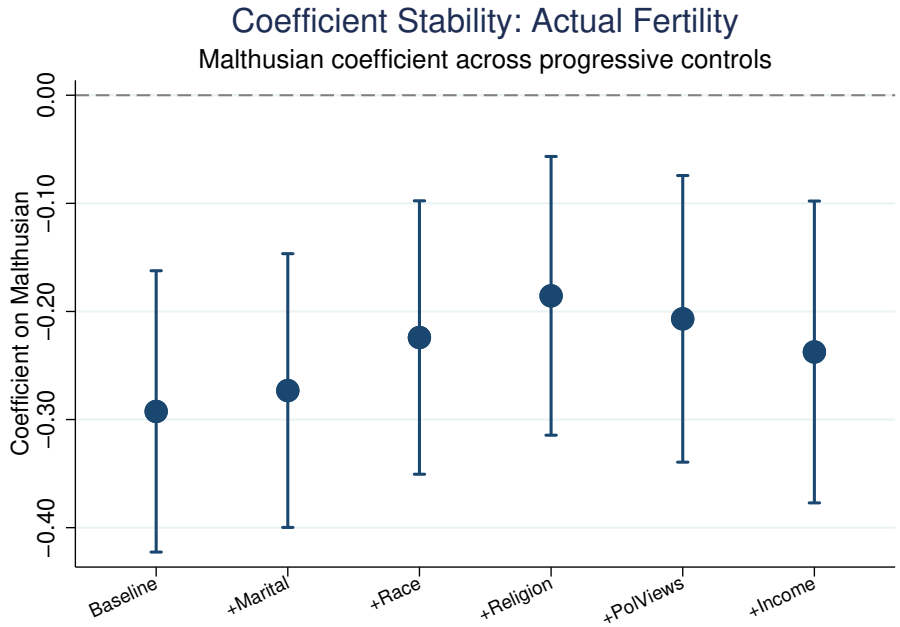


Figure A.3. Coefficient Stability Across Progressive Controls: Actual Fertility

Notes: Each point is the OLS coefficient on the Malthusian indicator from a regression of actual fertility with the indicated cumulative set of controls. The baseline includes age, age squared, sex, education, and year fixed effects. Error bars show 95% confidence intervals based on robust standard errors. N falls from 1,822 to 1,567 as controls with missing values are added. Table A.2 reports the corresponding ideal-fertility coefficients.

Table A.3. Alternative Specifications

	Count Models (Baseline)			Sensitivity	
	Ideal Poisson (1)	Actual Poisson (2)	Actual NegBin (3)	Ideal Strict (4)	Actual Strict (5)
Malthusian	-0.107^{***}	-0.179^{***}	-0.183^{***}		
	(0.016)	(0.037)	(0.037)		
Malthusian (excl. neutrals)				-0.357^{***}	-0.342^{***}
				(0.054)	(0.087)
N	1,822	1,822	1,822	1,412	1,412

Notes: Columns 1–3 report count-model coefficients with baseline controls (age, age squared, sex, education, year FE). Columns 4–5 use OLS with the same controls, restricting the sample to respondents who agree/disagree (excluding neutrals). Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

average marginal effect of -9.2 percentage points. With full controls the marginal effect is -8.0 percentage points. The intensive margin is marginally significant at baseline ($p = 0.089$) and

insignificant with full controls ($p = 0.285$). The association is therefore concentrated on the decision to become a parent.

Table A.4. Extensive and Intensive Margins of Actual Fertility

	Extensive: Any Children		Intensive: Children $\mid > 0$	
	Baseline (1)	Full (2)	Baseline (3)	Full (4)
Logit (odds ratios)				
Malthusian	0.554^{***}	0.533^{***}		
	(0.067)	(0.080)		
Average marginal effects				
Malthusian	-0.092^{***}	-0.080^{***}		
	(0.019)	(0.019)		
OLS				
Malthusian			-0.122^{*}	-0.085
			(0.072)	(0.080)
N	1,822	1,567	1,299	1,120

Notes: Columns 1–2 report logit odds ratios and average marginal effects for having any children. Columns 3–4 report OLS coefficients conditional on positive fertility. Baseline controls: age, age squared, sex, education, year FE. Full controls add marital status, race, religion, religiosity, political views, and log income. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure A.4 plots the share of respondents with any children across five levels of the Malthusian belief scale. The share declines from approximately 0.77 among those who disagree or are neutral to 0.60 among those who strongly agree, consistent with the logit estimates.

Cross-wave stability. Table A.5 estimates the baseline specification separately for the 2000 and 2010 waves. Both waves produce negative coefficients for both outcomes. The point estimates are larger in 2010, but pooled regressions with year–belief interactions fail to reject equality of the wave-specific coefficients ($p = 0.13$ for ideal, $p = 0.38$ for actual).

Distinguishing Malthusian beliefs from general environmentalism. Table A.6 adds a measure of general environmental concern (respondent believes too little is spent on the environment) to the regressions. The Malthusian coefficient is essentially unchanged when environmental concern is included, while environmental concern itself has only a marginal association with actual fertility and none with ideal fertility. This suggests that the Malthusian coefficient captures population-specific beliefs rather than broad environmental attitudes.

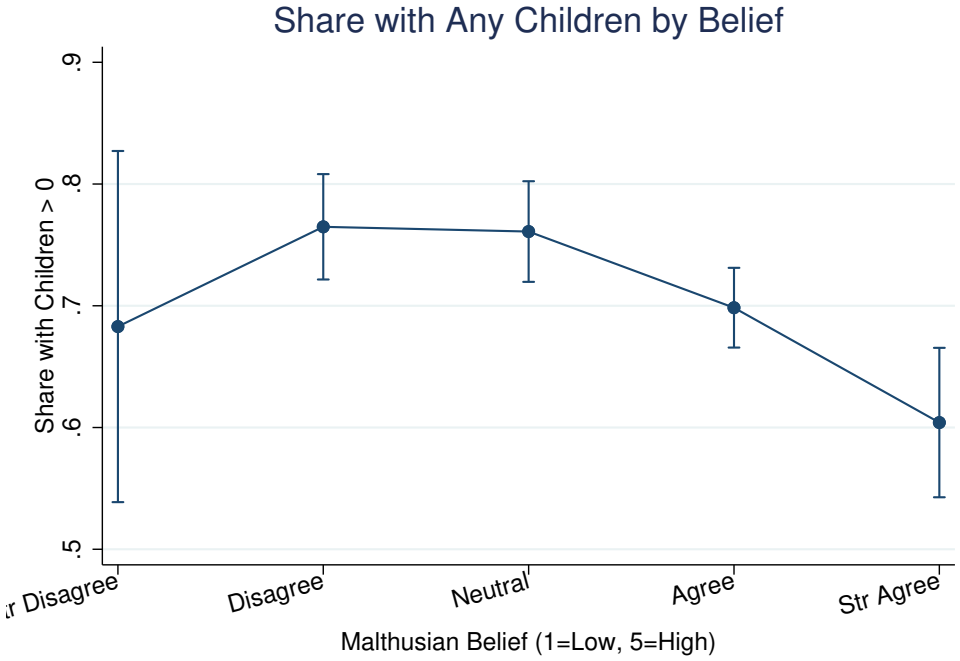


Figure A.4. Share with Any Children by Belief Intensity

Notes: Each point is the share of respondents reporting at least one child at the indicated level of the Malthusian belief scale ($N = 1{,}822$). Error bars show 95% confidence intervals. The "Strongly Disagree" category contains only 41 respondents, contributing to its wider interval.

Table A.5. Cross-Wave Stability

	Ideal Fertility		Actual Fertility	
	2000	2010	2000	2010
	(1)	(2)	(3)	(4)
Malthusian	-0.191^{***}	-0.313^{***}	-0.195^{*}	-0.350^{***}
	(0.062)	(0.053)	(0.111)	(0.083)
N	676	1,146	676	1,146
R^2	0.037	0.072	0.255	0.275

Notes: All models control for age, age squared, sex, and education. Robust standard errors in parentheses. Pooled regressions with Malthusian $\times$ Year 2010 interactions yield F-statistics of 2.26 ($p = 0.133$) for ideal and 0.76 ($p = 0.384$) for actual fertility. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The fertility gap. If Malthusian beliefs reduce ideal and actual fertility proportionally, the gap between them (ideal minus actual) should be unrelated to beliefs. OLS regressions of this gap on the Malthusian indicator yield coefficients of 0.024 ($p = 0.74$) with baseline controls and 0.035 ($p = 0.66$) with full controls. Among respondents with completed fertility the coefficient is -0.011

Table A.6. Malthusian Beliefs and General Environmental Concern

	Ideal Fertility		Actual Fertility	
	Baseline (1)	Full (2)	Baseline (3)	Full (4)
Malthusian	-0.288^{***}	-0.203^{***}	-0.248^{**}	-0.192^{*}
	(0.063)	(0.069)	(0.097)	(0.106)
Env. concern	-0.021	-0.021	-0.146^{*}	-0.143^{*}
	(0.048)	(0.053)	(0.075)	(0.081)
N	893	774	893	774
R^2	0.062	0.111	0.272	0.325

Notes: Environmental concern is coded so that higher values indicate greater concern. Baseline controls: age, age squared, sex, education, year FE. Full controls add marital status, race, religion, religiosity, political views, and log income. Sample is restricted to respondents with non-missing environmental attitudes. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

($p = 0.92$). The near-zero estimates are consistent with beliefs operating on both margins in similar proportions.

Reverse causality checks. Table A.7 addresses the concern that realized fertility shapes beliefs rather than the reverse. Column 1 reproduces the full-sample baseline for reference. Columns 2–4 split the sample by age group. For ideal fertility, the Malthusian coefficient is negative and significant in every group, including 18–34-year-olds who have had relatively few children. For actual fertility, the coefficient is essentially zero among 18–34-year-olds, largest among 35–49-year-olds, and significant among those 50 and older. This pattern, with no association when most childbearing lies ahead and a strong association among respondents old enough to have acted on their beliefs, is difficult to attribute to reverse causality. Column 5 adds the respondent's number of siblings to the baseline specification. Siblings are positively associated with both fertility measures, consistent with family-of-origin norms, but the Malthusian coefficient changes by less than five percent. A supplementary regression of the Malthusian indicator on siblings and demographic controls yields a coefficient of -0.008 ($p = 0.054$), indicating that family-of-origin size has at most a marginal direct association with population beliefs.

Completed fertility and heterogeneity.

Distributions of fertility by belief type. Figures A.5a and A.5b compare the full distributions of ideal and actual fertility for Malthusians and non-Malthusians. For ideal fertility, Malthusians

Table A.7. Reverse Causality Checks

	(1) Full Sample	(2) Age 18–34	(3) Age 35–49	(4) Age 50+	(5) +Siblings
Panel A: Ideal Fertility					
Malthusian	-0.268^{***}	-0.243^{***}	-0.322^{***}	-0.253^{***}	-0.256^{***}
	(0.041)	(0.075)	(0.079)	(0.061)	(0.040)
Siblings					0.044^{***}
					(0.008)
R^2	0.060	0.107	0.066	0.045	0.079
Panel B: Actual Fertility					
Malthusian	-0.292^{***}	-0.052	-0.469^{***}	-0.263^{**}	-0.283^{***}
	(0.066)	(0.094)	(0.124)	(0.114)	(0.066)
Siblings					0.041^{***}
					(0.014)
R^2	0.262	0.273	0.138	0.109	0.266
N	1,822	514	541	767	1,818

Notes: Column 1 reproduces the full-sample baseline (age, age squared, sex, education, year FE). Columns 2–4 estimate the same specification separately by age group. Column 5 adds the respondent's number of siblings to the baseline. Sample size in column 5 is four fewer due to missing sibling data. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.8. Actual Fertility by Completed Fertility Status

	Full Sample (1)	Completed Fertility (2)	Not Yet Completed (3)
Malthusian	-0.292^{***}	-0.257^{**}	-0.288^{***}
	(0.066)	(0.109)	(0.079)
Observations	1,822	848	974
R^2	0.262	0.112	0.266

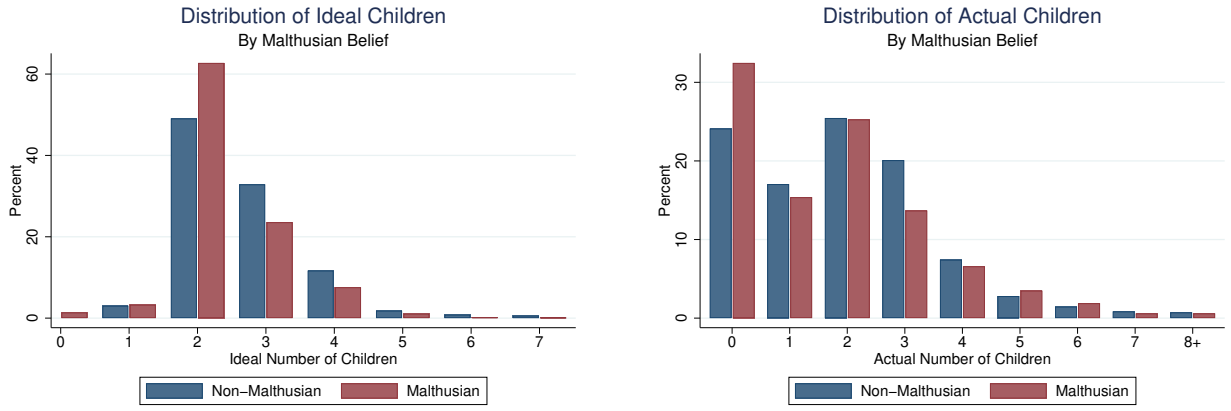
Notes: he dependent variable is actual number of children. Completed fertility denotes women aged 45+ and men aged 50+. All models control for age, age squared, sex, education, and year. Robust standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

are more concentrated at two children and less likely to report three or more. For actual fertility, Malthusians are more likely to have zero children, consistent with the extensive-margin result above.

Table A.9. Heterogeneity in the Belief–Fertility Relationship

	Ideal Fertility		Actual Fertility	
	HS or Less	Some College+	HS or Less	Some College+
Malthusian	-0.318^{***}	-0.235^{***}	-0.209^{*}	-0.392^{***}
	(0.069)	(0.049)	(0.115)	(0.081)
Observations	821	1,001	821	1,001
R^2	0.045	0.029	0.163	0.240

Notes: The dependent variable is ideal fertility in columns 1–2 and actual fertility in columns 3–4. Robust standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



(a) Ideal children

(b) Actual children

Figure A.5. Distribution of Fertility by Malthusian Belief

Notes: Bars show the percentage of respondents reporting each number of children, separately for non-Malthusians and Malthusians ($N = 1{,}822$).

D Quantitative Appendix

This appendix supplements Section 4 with computational details, the welfare construction, the benchmark-comparison paths, the modern fertility-decline calibration, sampling uncertainty, consumption comparisons, and parameter sensitivity.

D.1 Computation and Calibration Provenance

The simulations iterate the exact laws of motion for the gap (2), the social credence (A.18), and the composition (8). Fertility policies implement the first-order response (A.12) in log form,

$$\ln n_{j,t} = \nu + K\left[\bar{\theta}_p X_t^1 + (\theta_j - \bar{\theta}_p) X_t^{\lambda}\right],$$

with the average belief $\bar{\theta}_p$ evaluated at the current credence, $p = \pi_t$, the invariant row of the child-type transition matrix (A.13), and with $\nu = 0$ in the historical exercises. The log form guarantees positive fertility and coincides with the paper's expansion to first order. The discounted gaps X_t^λ and X_t^1 are evaluated with the constant-fertility forecast rule of Lemma 2 at $m^e = \ln \bar{n}_t$, and aggregate fertility is then determined as a within-period fixed point of $\bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}$. The fixed-point map is strongly contracting, with a local slope of roughly 0.07, and is solved to a tolerance of 10^{-12}. Fertility is restricted to $\mathcal{N} = [0.3, 3.0]$, a TFR range of 0.6 to 6 lying strictly inside the feasibility bound $n_{\max} \simeq 3.58$ of Assumption 1; paths that reach the upper bound are reported at the cap, and beliefs are kept inside $(0, 1)$ by a 10^{-9} bound. Deterministic figures set the productivity innovations to zero, so growth observations equal their conditional means $\theta^* x_t$; the Monte Carlo exercise draws $\eta_{t+1} \sim \mathcal{N}(0, \sigma^2)$ independently across periods and paths. The horizon is 80 generations, or 2,000 years, with entry evaluated at $t = 0, \ldots, 80$.

The preference parameters follow the illustrations in Section 2 and Appendix A.2: $\rho = 0.5$ within the required interval $(0, 1)$, $\varepsilon = 0.5$ consistent with the curvature restrictions discussed by Jones and Schoonbroodt (2010) for Barro–Becker dynasties, $\gamma \simeq 0.3122$ from a 15 percent rearing-time share at replacement, and $\beta = \beta^\dagger \simeq 0.2517$ from the replacement normalization, so that β is pinned by the normalization rather than calibrated freely. The implied constants are $\delta = 0.2530$, $K = 1/3$, $\bar{\theta}^{dest} = 0.2241$, $p^* = 0.5756$, and $a_C = 1.1337$.

The least directly disciplined parameter is μ. The value 0.10 per generation implies a gap half-life of about 6.6 generations, or 165 years, under replacement fertility, which is consistent with the very slow transition dynamics implied by semi-endogenous growth models (Jones, 1995, 2022). It also ensures $\bar{\theta}^{dest} < \theta_R$, so that the premise $a_C > 1$ of Proposition 4 holds. The strong scale effect $\theta_R = 0.5$ targets the historical takeoff question: the implied long-run multiplier $\theta_R/\mu = 5$ corresponds to the powerful population–technology feedback that Kremer (1993) documents over the very long run, and the entry threshold $h = 1$ then delivers expected growth of $\theta_R h = 0.5$ per generation, about two percent per year, at entry. By contrast, mapping modern semi-endogenous estimates into the gap specification, using the idea-production parameters in Jones (2022) and the declining research productivity documented by Bloom et al. (2020), yields a scale effect on the order of $0.3\mu \simeq 0.03$, well below $\bar{\theta}^{dest}$. Under such values even the correct-belief economy would not amplify a displacement, and the economy would sit in the dormant region regardless of beliefs. The baseline should therefore be read as a calibration of the takeoff-era feedback regime, with the modern-estimate case noted as a robustness remark rather than an alternative baseline.

D.2 The Welfare Measure

The welfare cost ω in Section 4.3 is a dynastic consumption-equivalent measure over a truncated window of $T_w = 20$ generations. For each economy $k \in \{H, C\}$, window utility is computed by

the backward recursion

$$V_t^k = \bar{f}_t^k + \beta^\dagger \left(\bar{n}_t^k\right)^{1-\varepsilon} \left(\frac{A_{t+1}^k}{A_t^k}\right)^{1-\rho} V_{t+1}^k, \qquad V_{T_w}^k = \bar{f}_{T_w}^k,$$

where $\bar{f}_t = \phi_t f(n_{R,t}) + (1-\phi_t) f(n_{M,t})$ is the population-weighted flow utility (A.2), so V_0^k is the population-average dynastic utility of an initial adult along the equilibrium path. Because values are homogeneous of degree $1-\rho$ in consumption, the uniform proportional consumption supplement that equates the two window utilities is

$$\omega = \left(\frac{V_0^C}{V_0^H}\right)^{1/(1-\rho)} - 1.$$

The window is truncated for two reasons. First, the fertility cap binds in the correct-belief economy from generation 23 onward, so later comparisons mix the belief mechanism with the mechanical effect of the cap. Second, once takeoff produces sustained growth, the effective dynastic growth factor $\beta^\dagger \bar{n}^{1-\varepsilon} e^{(1-\rho)\Delta \ln A}$ exceeds one, so the untruncated dynastic sum diverges and an infinite-horizon consumption equivalent is undefined in that region.

The benchmark value is $\omega = 0.74$ percent of consumption per generation; across the $\lambda \times \sigma$ grid of Table A.11, the range is 0.71 to 0.75 percent. Two caveats apply. First, the measure compares dynastic utilities per initial adult and holds aside the valuation of population size itself, an issue studied by Golosov et al. (2007), Blackorby et al. (2005), and, in a measurement context, Adhami et al. (2026); a criterion that values additional lives would weigh the delayed expansion differently. Second, with an effective dynastic discount factor of $\delta \simeq 0.25$ per generation, losses accruing beyond a few generations receive little weight, while the undiscounted consumption differences become very large (Figure A.8). The reported ω is therefore a conservative, preference-consistent valuation of the entry delay.

D.3 The Correct-Belief Benchmark Comparison

Figure A.6 displays the population-gap paths for the inherited-disagreement economy H and the correct-belief economy C of Proposition 4, from a common displacement $x_0 = 0.30$. The two economies share the technology, the Romerian truth, and the displacement; they differ only in beliefs. The informed economy amplifies from the first generation and takes off after 10 generations. The inherited-disagreement economy first contracts the gap and takes off 45 generations later. The 1,125-year entry delay is attributable entirely to inherited beliefs.

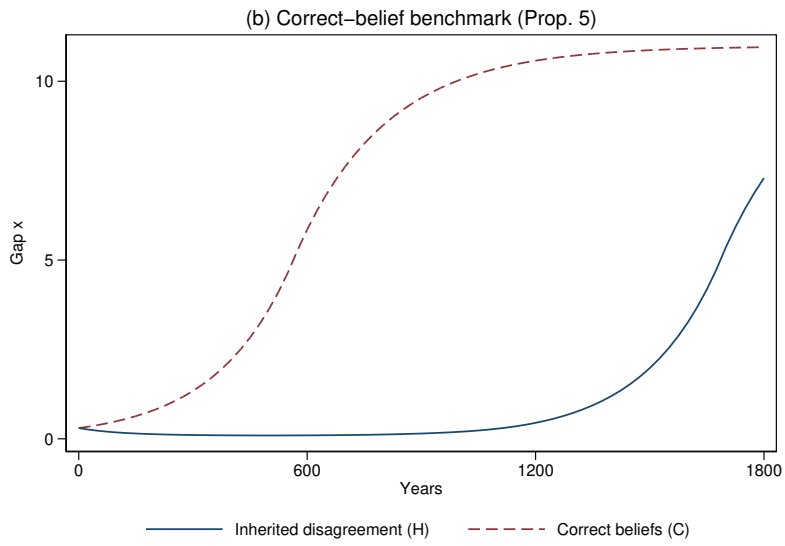


Figure A.6. Takeoff Delay and the Correct-Belief Benchmark: Full Comparison

Notes: population-gap paths for the inherited-disagreement economy H and the correct-belief economy C of Proposition 4, from a common displacement $x_0 = 0.30$.

D.4 Modern Fertility Decline: Quantitative Illustration

The modern application recalibrates three objects to a 2025 starting point and retains everything else. A common preference shift $\nu = \ln(0.875)$ sets zero-gap fertility to a TFR of 1.75, between the current U.S. period rate and the completed fertility of recent cohorts (Doepke et al., 2023). The initial Malthusian share is 0.55, taken directly from the GSS (Table A.1). The perceived population gap of 0.48 is chosen so that the model's equilibrium fertility differential between types matches the GSS fertility gap: the model delivers a TFR difference of 0.21, close to the full-controls actual-fertility estimate of 0.24 in Table 3 (0.29 at baseline). Appendix D.1 details this identification, which operates through Lemma 2 and equation (13).

Figure A.7 reports the projection against a counterfactual that holds both the adult composition and the social credence at their 2025 values while the perceived gap continues to evolve. With a positive discounted expected gap, Romerians have the reproductive advantage, and the Malthusian share falls from 0.55 to 0.42 by 2050 and 0.20 by 2125. Through 2075, selection raises aggregate fertility by about 0.021 children relative to the frozen composition. By 2100 the effect is near zero, and by 2125 it turns negative, at -0.022, as the perceived gap approaches and crosses the reference. Belief selection therefore first cushions and later deepens the fertility decline, making the reversal implied by Proposition 1 quantitatively visible within a single century.

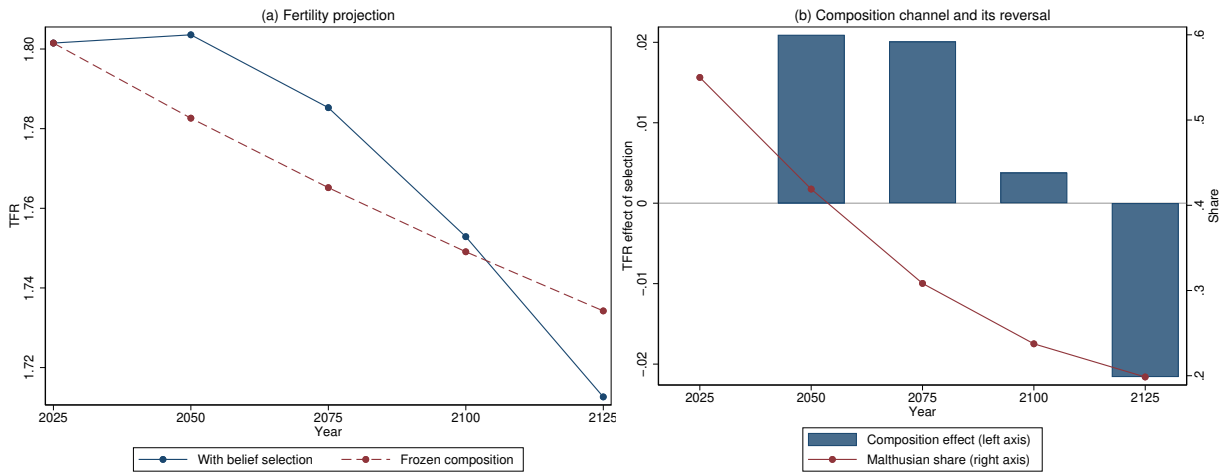


Figure A.7. Belief Selection and Modern Fertility Decline

Notes: Calibration to 2025: zero-gap TFR of 1.75 ($\nu = \ln 0.875$), initial Malthusian share of 0.55 from the GSS, and a perceived gap of 0.48 matched to the GSS fertility differential. Panel (a) compares the equilibrium TFR path with a counterfactual holding both the adult composition and the social credence at their 2025 values while the perceived gap continues to evolve. Panel (b) plots the difference between the two paths (bars, left axis) and the Malthusian population share (line, right axis).

D.5 Consumption Paths in the Benchmark Comparison

Consumption is measured as $A_t(1 - \psi(\bar{n}_t))$, the wage net of the child-rearing time cost evaluated at aggregate fertility. Figure A.8 plots the log-consumption difference between the inherited-disagreement and correct-belief economies of Figure A.6. The difference is positive for roughly the first two generations: disagreement lowers fertility, which frees parental time and raises consumption while productivity paths remain close. It turns negative thereafter and widens steadily, reaching about 10 log points by generation 20, the end of the welfare window, and growing far larger beyond. Levels beyond the takeoff of both economies reflect the fertility cap and are illustrative only; they are the reason the welfare window in Appendix D.2 stops at generation 20.

D.6 Sensitivity and Robustness

Table A.10 varies one parameter at a time around the benchmark and reports the takeoff delay relative to the correct-belief economy. The patterns follow the mechanism. Raising the vertical transmission probability lengthens the delay, from 950 years at $\lambda = 0.4$ to 1,125 at the baseline 0.6 and beyond the simulation horizon at 0.8, because stronger family transmission insulates the composition from the improving social credence; in the limit of Theorem 2, composition responds to evidence only through births. Lowering the signal noise shortens the delay dramatically, to 175 years at $\sigma = 0.1$, while $\sigma = 0.3$ is censored: information per observation is proportional to $1/\sigma^2$, so volatility directly governs how long inherited beliefs survive contact with the data. Varying the

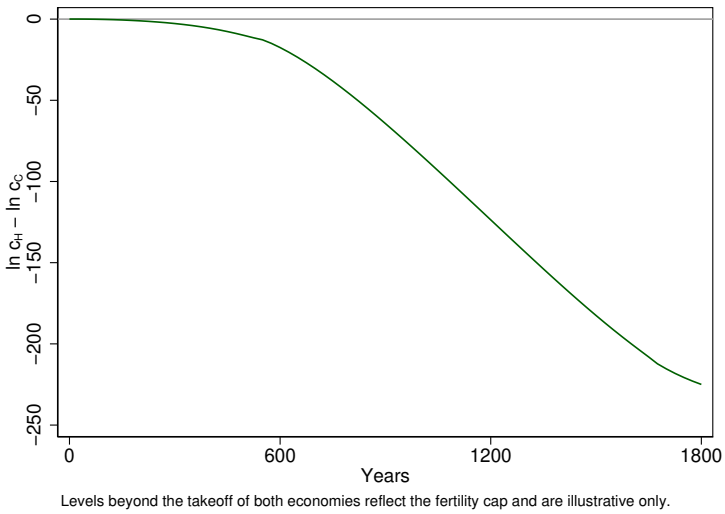


Figure A.8. Log-Consumption Difference: Inherited Disagreement Minus Correct Beliefs

Notes: Difference in log consumption between economies H and C of Proposition 4 under the baseline calibration. Consumption is $A_t(1-\psi(\bar{n}_t))$. Levels beyond the takeoff of both economies reflect the fertility cap and are illustrative only.

Malthusian parameter between -0.075 and -0.30 moves the delay only between 975 and 1,225 years. Greater pessimism strengthens the contraction of the gap, which reduces information, but it also widens $\Delta\theta$, which increases the information content of any given gap and strengthens selection; the two effects largely offset.

Table A.10. Takeoff Delay: One-Parameter Sensitivity

Parameter	Value	Delay (years)
λ	0.400	950
λ	0.600	1,125
λ	0.800	>1,775
σ	0.100	175
σ	0.200	1,125
σ	0.300	>1,775
θ_M	-0.075	975
θ_M	-0.150	1,125
θ_M	-0.300	1,225

Notes: Takeoff delay relative to the correct-belief economy, varying one parameter around the benchmark ($\lambda = 0.6$, $\sigma = 0.2$, $\theta_M = -0.15$) with all other parameters at their Table 2 values. Entries of >1,775 indicate no takeoff within the 2,000-year simulation horizon.

Table A.11 reports the full $\lambda \times \sigma$ grid together with the welfare cost ω. The delay interacts strongly with both parameters: high transmission and high noise are jointly sufficient for censoring,

while low noise produces takeoff within a few generations even at high transmission. The welfare cost is remarkably stable, between 0.71 and 0.75 percent per generation, across all nine cells, including the censored ones. This stability reflects the discounting logic of Appendix D.2: the window utilities are dominated by the early generations, in which all configurations share a similar contraction phase, so ω measures the common early cost of pessimistic fertility restraint rather than the full cost of the eventual delay.

Table A.11. Robustness: Delay and Welfare Cost Across λ and σ

λ	σ	Delay (years)	ω (%)
0.40	0.10	150	0.72
0.40	0.20	950	0.74
0.40	0.30	>1,775	0.71
0.60	0.10	175	0.73
0.60	0.20	1,125	0.74
0.60	0.30	>1,775	0.71
0.80	0.10	275	0.75
0.80	0.20	>1,775	0.75
0.80	0.30	>1,775	0.71

Notes: Takeoff delay relative to the correct-belief economy and welfare cost ω (percent of consumption per generation over the 20-generation window of Appendix D.2) for each (λ, σ) pair, with all other parameters at their Table 2 values. Entries of >1,775 indicate no takeoff within the 2,000-year simulation horizon.

E Notation and Conditions

Table A.12. Core Notation

Symbol	Meaning	Defined
A_t	Productivity	(4)
N_t	Adult population	(1)
x_t	Population relative to its adjusting reference	(2)
μ	Reference catch-up rate	(2)
θ_R, θ_M	Rival scale-effect parameters	(4)
θ^*	True scale effect	(4)
$\Delta\theta$	Belief gap, $\theta_R - \theta_M$	(4)
ρ	CRRA parameter	Sec. 2
$\varepsilon, \gamma, \beta$	Dynastic curvature, child-cost, and altruism parameters	Sec. 2
σ^2	Productivity-shock variance	(4)
ϕ_t	Adult Romerian population share	(1)
π_t	Romerian social credence	(A.18)
λ	Probability of parental transmission	(A.3)
P_{jk}	Child-type transition probability	(A.3)
$\psi, f, n_{\max}$	Time cost, net flow utility, and fertility bound	(A.1)–(A.2)
C_j	Normalized expected child value	(A.5)
$\mathcal{I}_t$	Information in one growth observation	(7)
$\mathcal{C}_T$	Cumulative information	(21)
κ	Reference utility-scale multiplier	(A.7)
δ	Effective reference discount factor	(A.7)
$\beta^\dagger$	Replacement-normalized altruism weight	(A.9)
w^0	Common normalized reference value	(A.10)
$K, \bar{\theta}_p$	Fertility-response constant and average belief	(17)
X_t^λ	Discounted gap for the fertility difference	(12)
X_t^1	Discounted gap for average fertility	(A.11)
$a(p)$	Local gap coefficient	(15)
$b(p)$	Aggregate log-fertility slope, $a(p) - (1 - \mu)$	(A.26)
B_t	Reproductive advantage (birth minus adult share)	(A.19)
T_h^H, T_h^C	Inherited- and correct-belief entry times	(22)
ν	Perturbation in reference log fertility	(A.45)
$D(p)$	Net stationary adjustment coefficient	Prop. 6
ω	Window consumption-equivalent welfare cost	App. D.2

Table A.13. Conditions and Their Roles

Object	Condition	Role
Household value	Assumption 1	Finite normalized continuation value
Reference normalization	$\beta = \beta^{\dagger}$	Replacement fertility on the zero-gap path
Local equilibrium analysis	Assumption 2 and $\delta\|a(p)\| < 1$	Smooth, discounted derivative continuation
Uniform local return	Assumption 3	High-probability geometric contraction
Damped demographic cycles	Uniform negative response, (A.40)	Alternating, shrinking gaps
Truth identification	$\sum_t \mathcal{I}_t = \infty$	Posterior probability of truth converges to one
Persistent disagreement	$\sum_t \mathcal{I}_t < \infty$	Interior posterior limit
Entry-delay comparison	Inherited-belief contraction and $a_C > 1$	Different responses to a common displacement
Stationary comparison	Fixed beliefs and $D(p) > 0$	Regular local stationary adjustment

Notes: Local coefficient restrictions apply to the maintained smooth equilibrium branch. Global equilibrium existence requires additional analysis. The four-phase diagram organizes a conditional adjustment sequence, while the damped-cycle corollary gives a local result. The fixed-belief stationary comparison suspends learning and selection. The baseline common log-growth intercept is zero.